What is sodium fusion test:

⇒ The sodium fusion test is used in elemental analysis for the quantitative determination of elemental halogen, nitrogen and sulfur in a sample. It is developed by J.L. Lassaigne.

$$C, N, S, X \xrightarrow{Na,\ heat} Na_2S, NaCN, NaCNS, NaX$$

[X = Cl, Br and I]

Detection of Sulfur:

(i) Lead acetate test:

$$Na_2S + (CH_3COO)_2Pb \xrightarrow{dil\ CH_3COOH} PbS\downarrow + 2CH_3COONa$$

(black or brown ppt)

(a) Firstly, the fusion filtrate is acidified with dil CH_3COOH until the solution is acidic. Then if we add lead acetate, the solution containing the sulfur react to give the black precipitate of PbS which indicates the presence of sulfur.

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(ii) Sodium Nitroprusside test: If we add only a drop of 0.1% Sodium nitroprusside to the fusion filtrate, if the filtrate contains sulfur, the reaction will ~~contain~~ produce a deep red or purple color solution.

$$Na_2S_{(aq)} + Na_2[Fe(CN)_5NO]_{(aq)} \rightarrow Na_4[Fe(CN)_5NOS]$$

(purple color)

Detection of Nitrogen:

(i) Prussian Blue test:

$$NaOH + FeSO_4 \longrightarrow Fe(OH)_2 + Na_2SO_4$$

$$Fe(OH)_2 + NaCN \longrightarrow Na_4[Fe(CN)_6] + 2NaOH$$

Sodium ferrocyanide (blue)

$$3Na_4[Fe(CN)_6] + 4FeCl_3 \longrightarrow Fe_4[Fe(CN)_6]_3 + 12NaCl$$

Ferric ferro cyanide (Prussian blue)

The prussian blue color or blue precipitate indicates the presence of Nitrogen.

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⊞ How would you give a combine test for N and S?

⇒ If any organic compound contains N and S during the sodium fusion it may produce sodium thiocyanate (NaCNS). A fusion filtrate containing NaCNS is first acidified with dil HCl, then a drop of freshly prepared 2.5% $FeCl_3$ will give a blood red coloration due to the formation of the compound ferric thiocyanochloride which indicates the simultaneous presence of N and S.

$$NaCNS + FeCl_3 \rightleftharpoons Fe(CNS)Cl_2 + NaCl$$

Ferric thiocyanochloride (blood color)

⊞ Detection of X: If N and S is present add HNO_3 to acidify the solution filtrate by heating the reduce halide

$$NaCN + HNO_3 \xrightarrow{\Delta} NaNO_3 + HCN\uparrow$$

$$Na_2S + HNO_3 \xrightarrow{\Delta} NaNO_3 + H_2S\uparrow$$

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For Cl

$$NaCl + AgNO_3 \longrightarrow AgCl\downarrow + NaNO_3$$
(white)

$$NaBr + AgNO_3 \longrightarrow AgBr\downarrow + NaNO_3$$
(pale yellow)

$$NaI + AgNO_3 \longrightarrow AgI\downarrow + NaNO_3$$
(yellow)

Differentiation of Cl, Br, I:

$$NaCl + HNO_2 \xrightarrow{CCl_4} NaNO_3 + HCl$$

$$HCl + HNO_3 \longrightarrow Cl_2 \text{ (in } CCl_4\text{)}$$
(colorless)

$$NaBr + HNO_2 \xrightarrow{CCl_4} NaNO_3 + HBr$$

$$HBr + HNO_3 \longrightarrow Br_2 \text{ (in } CCl_4\text{)}$$
(Deep brown)

$$NaI + HNO_3 \xrightarrow{CCl_4} NaNO_3 + HI$$

$$HI + HNO_3 \longrightarrow I_2 \text{ (in } CCl_4\text{)}$$
(violet)

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⊞ Amines test–

⊞ Test for primary aliphatic amins:

(1) Nitrous acid test:

$10\%\ NaNO_2 + HCl \xrightarrow[0-5^{\circ}C]{} HNO_2$

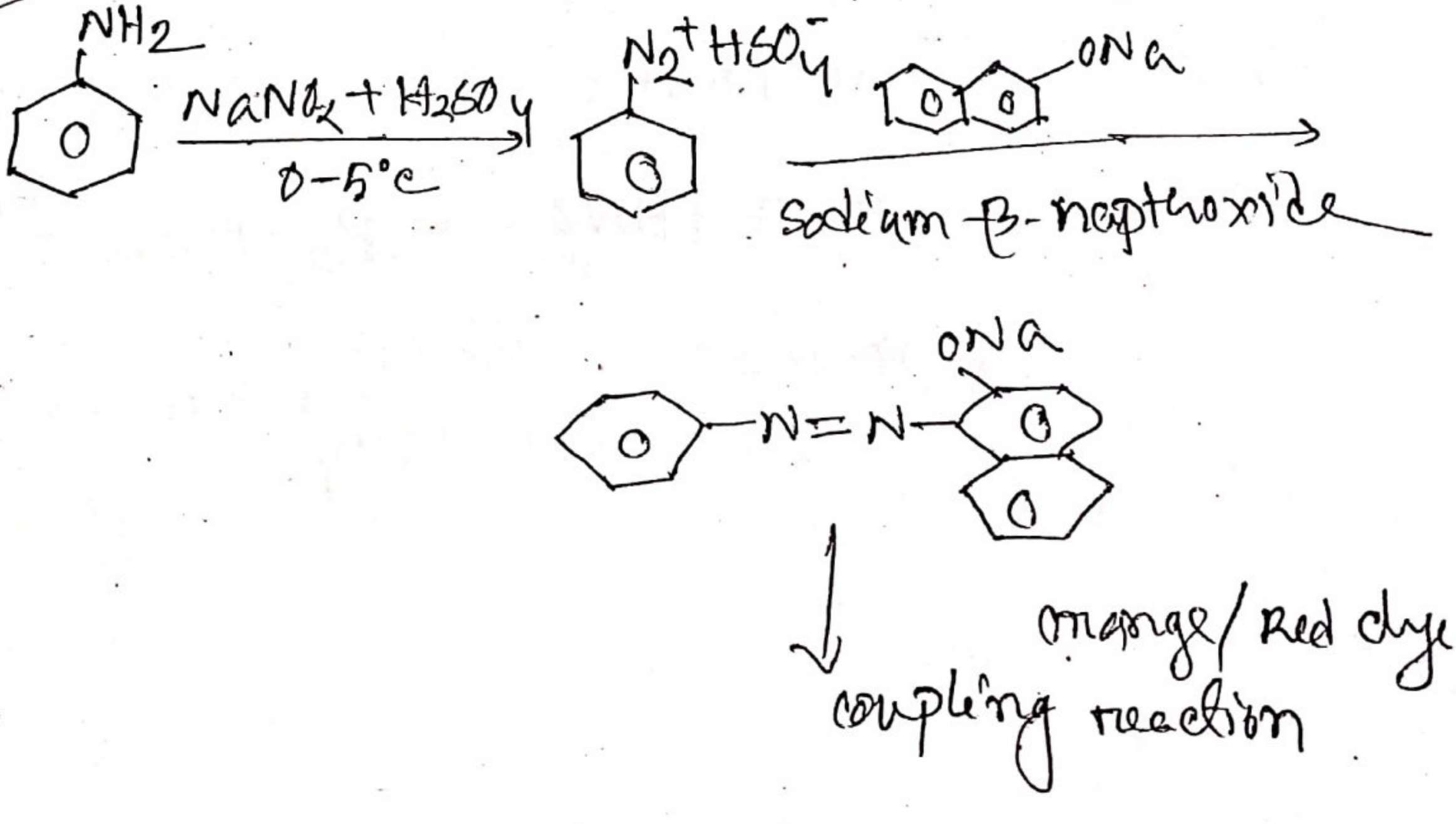
$HNO_2 + R-NH_2 \longrightarrow R-\overset{\oplus}{N}-N=O$

(Oily dark colour liquid)

⊞ Test for primary aromatic amin:

① Diazolization test:

T-10

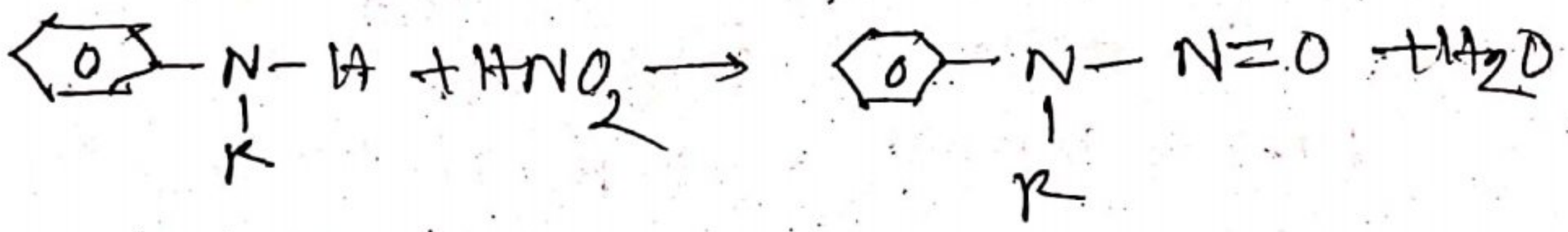


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Test for 2° amine

(i) Nitrous acid test:

$$R_2NH + HNO_2 \longrightarrow R_2N{-}N{=}O + H_2O$$

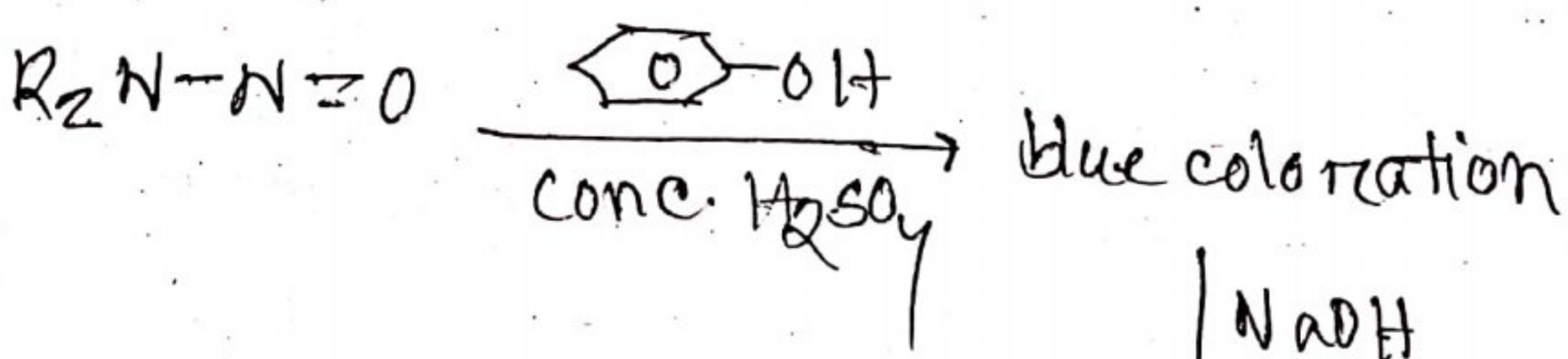


oily dark colored liquid indicates the presence of secondary amine.

(ii) Liebermann test:

$$HCl + NaNO_2 \longrightarrow HNO_2 + NaCl$$

$$R_2NH + HNO_2 \longrightarrow R_2{-}N{=}N{=}O + H_2O.$$



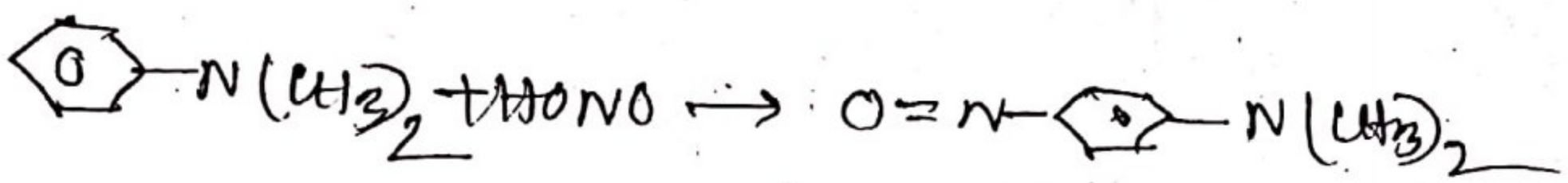
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Tertiary amine test.

Aliphatic:

$$(CH_3)_3N + HO-NO \rightarrow [(CH_3)_3N^+H]\ NO_2^-$$

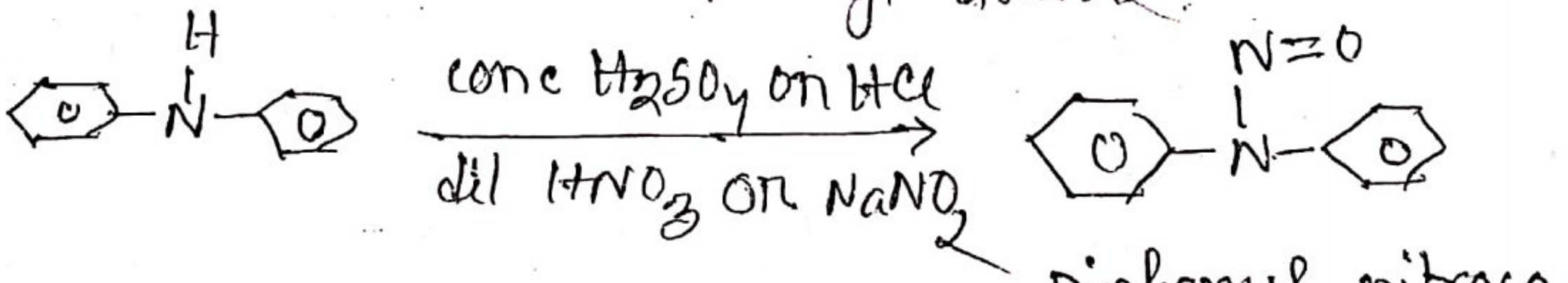
Aromatic:



P-nitroso dimethyl aniline

No reaction indicates aliphatic tertiary amine. Production of green or brown colored salt indicates the presence of aromatic tertiary amine.

Special test for diphenyl amine:

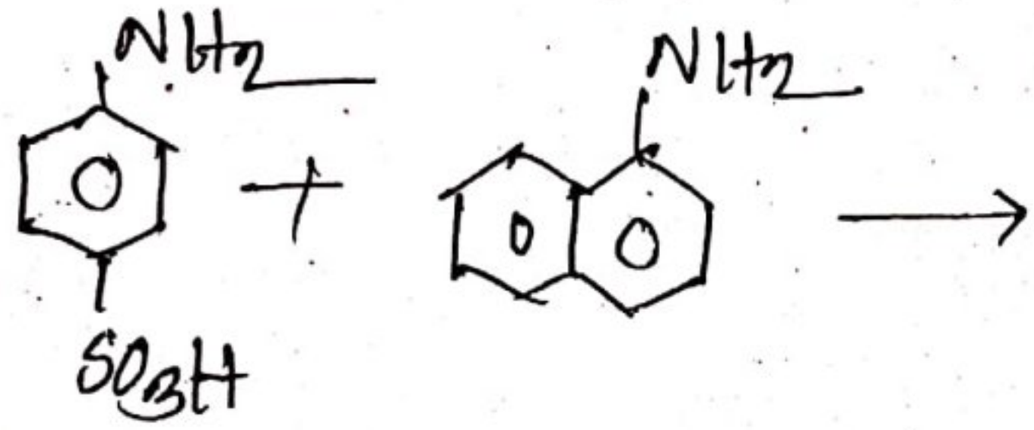


Diphenyl nitroso amine

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⊞ Nitro compound test

Griess Reagent: Griess reagent is prepared by using equal volume of 1.0% Sulphanilic acid solution and 0.1% 1-naphtyl amine.



⊞ Special test for di and trinitro Benzene.

⊞ General test for amide

1) Nitrus acid test.

$$R-CO-NH_2 + HO-NO \longrightarrow R-\overset{\overset{O}{\|}}{C}-OH + N_2 + H_2O$$

(Brisk effervesence due to evolution of N_2 gas)

⊞ Special test for aliphatic amide.

$$R-\overset{\overset{O}{\|}}{C}-NH_2 + H_2N-OH \rightarrow R-\overset{\overset{O}{\|}}{C}-NH-OH + NH_3$$

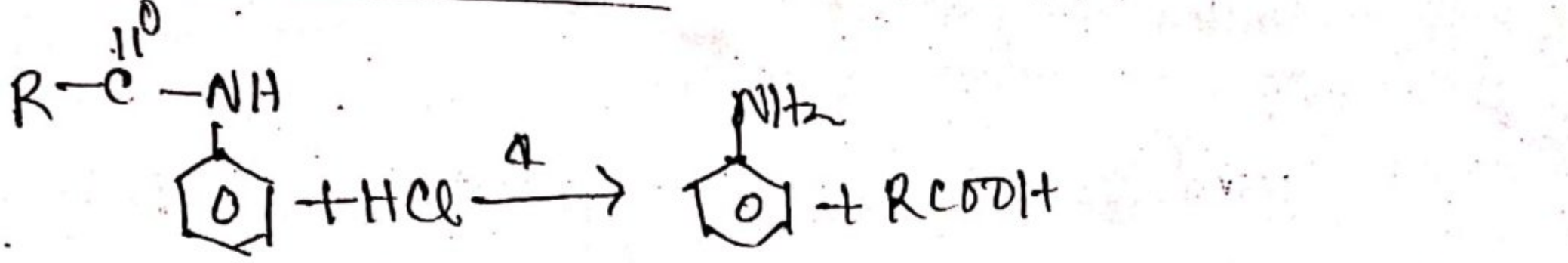
$$R-\overset{\overset{O}{\|}}{C}-NH-OH + FeCl_3 \rightarrow \left[R-C(=O)-N(H)-O\right]_3 Fe$$

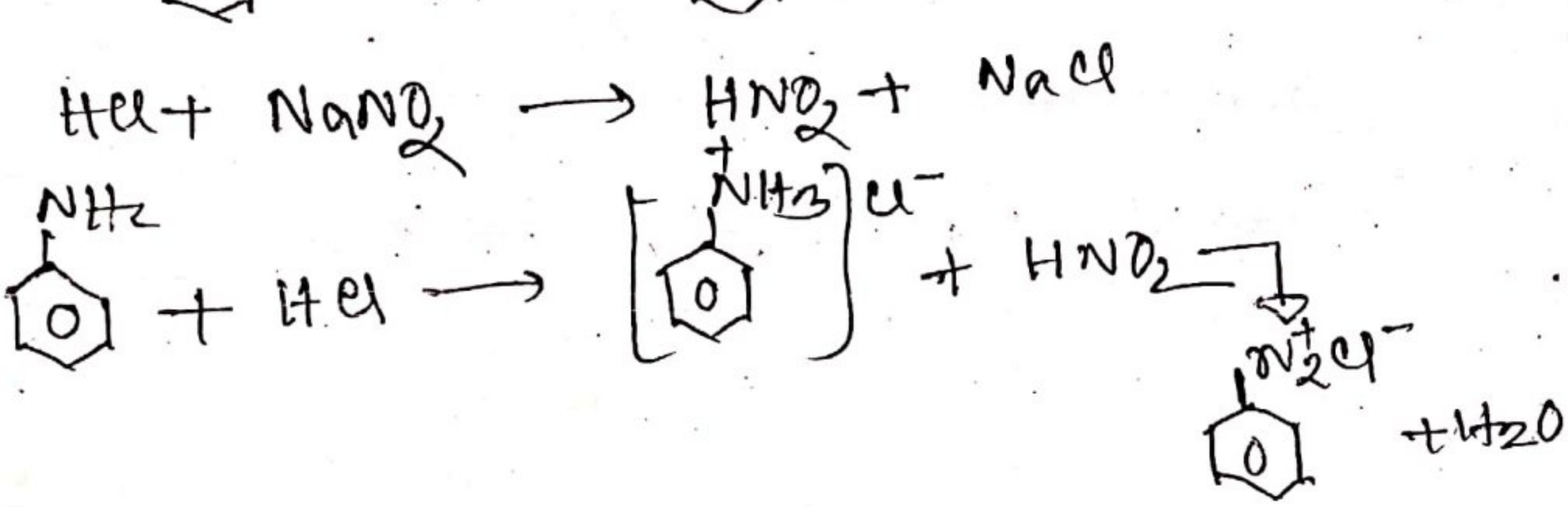
Blue red.

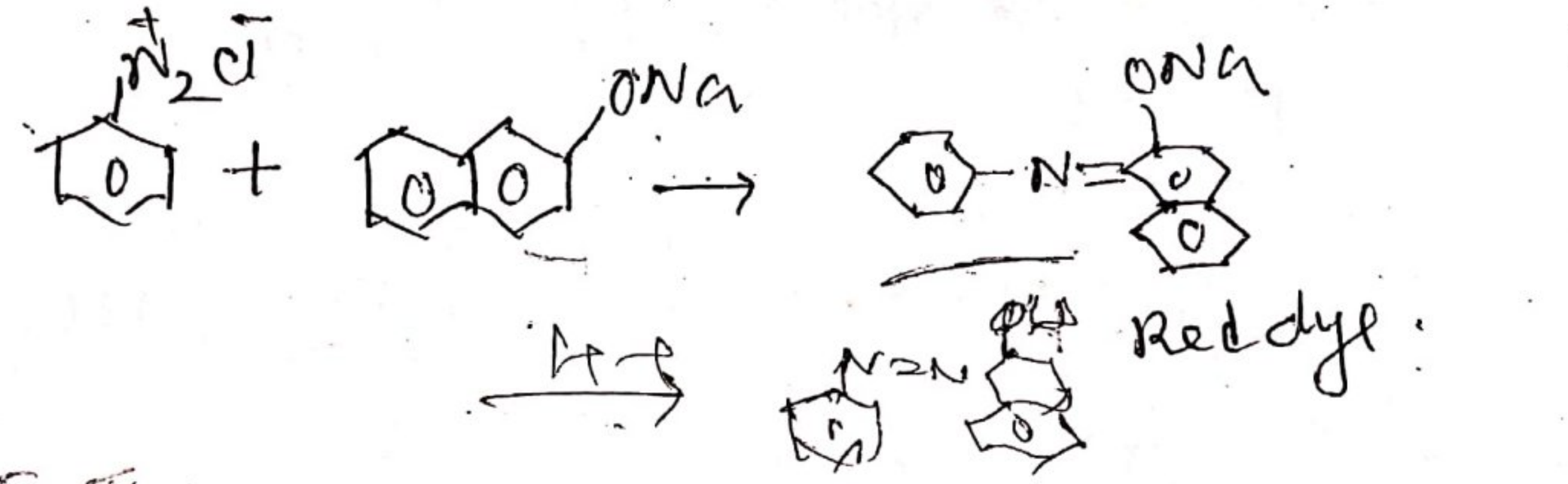
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Test for amide

① Diazotization test:

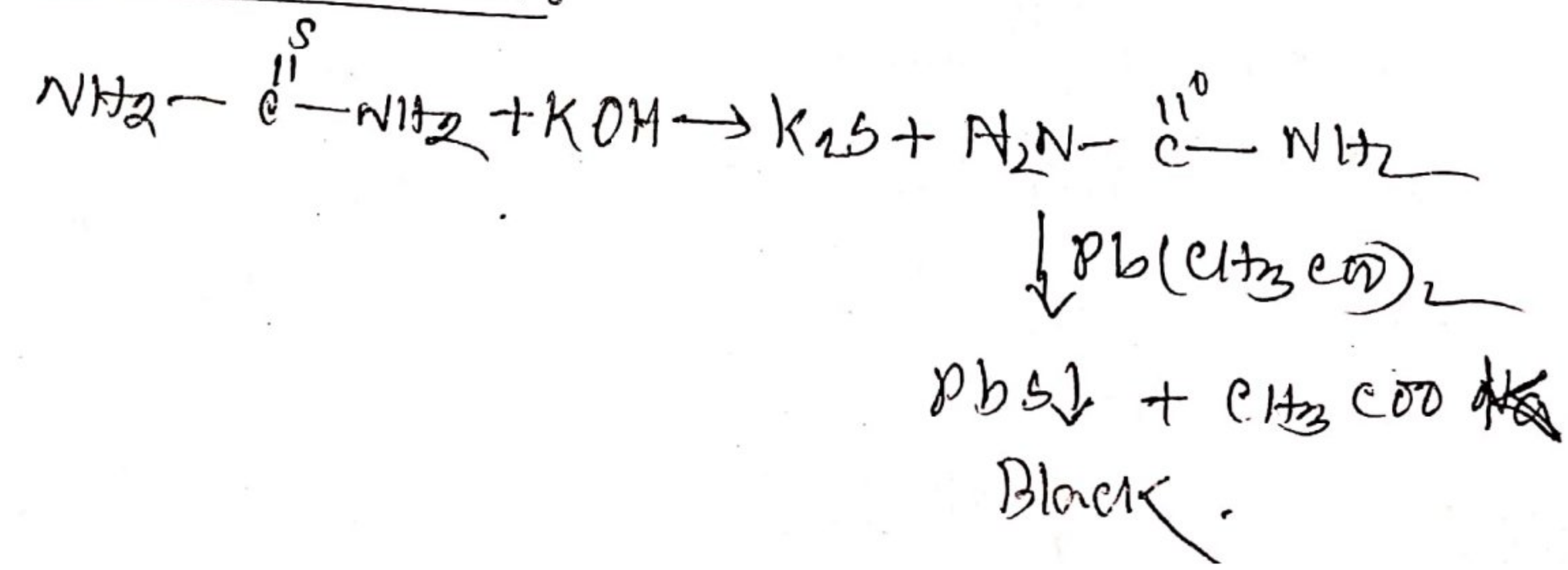






Thiourea test—

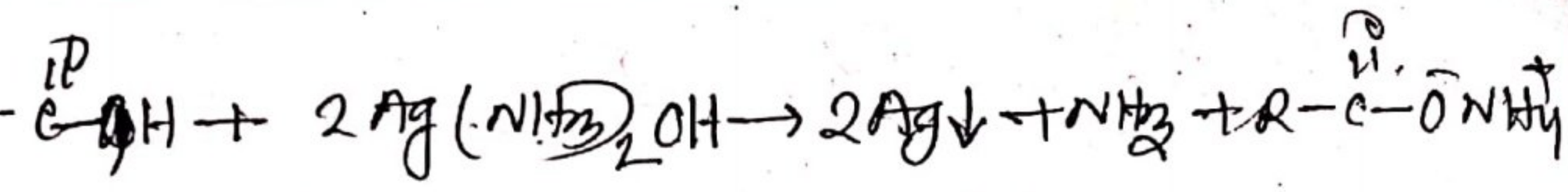
lead acetate test:



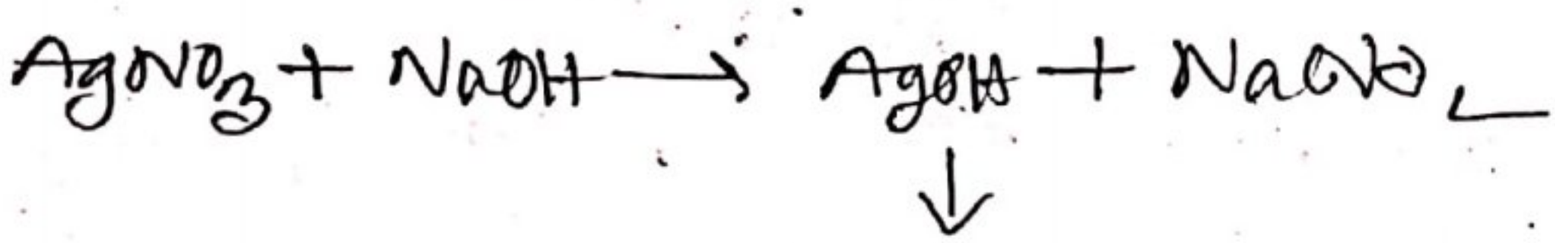
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⊞ Test for aldehyde

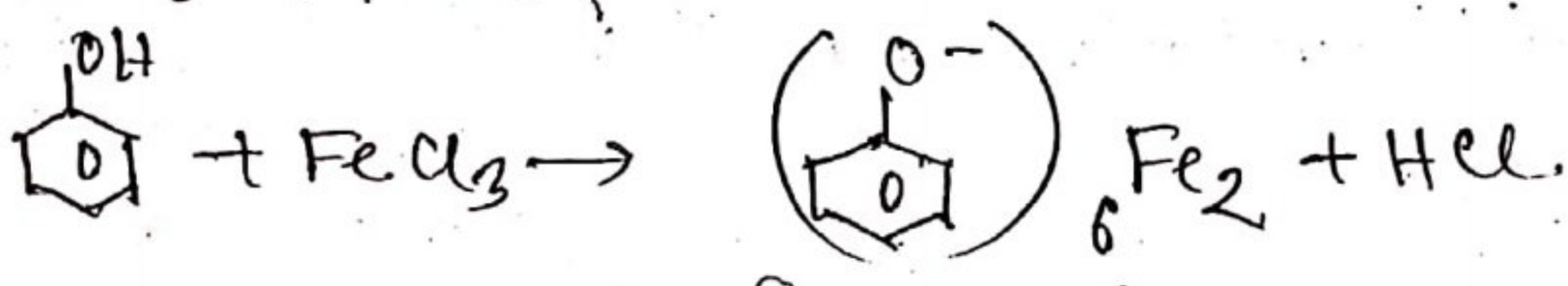
Tollen reagent test // oxidation test



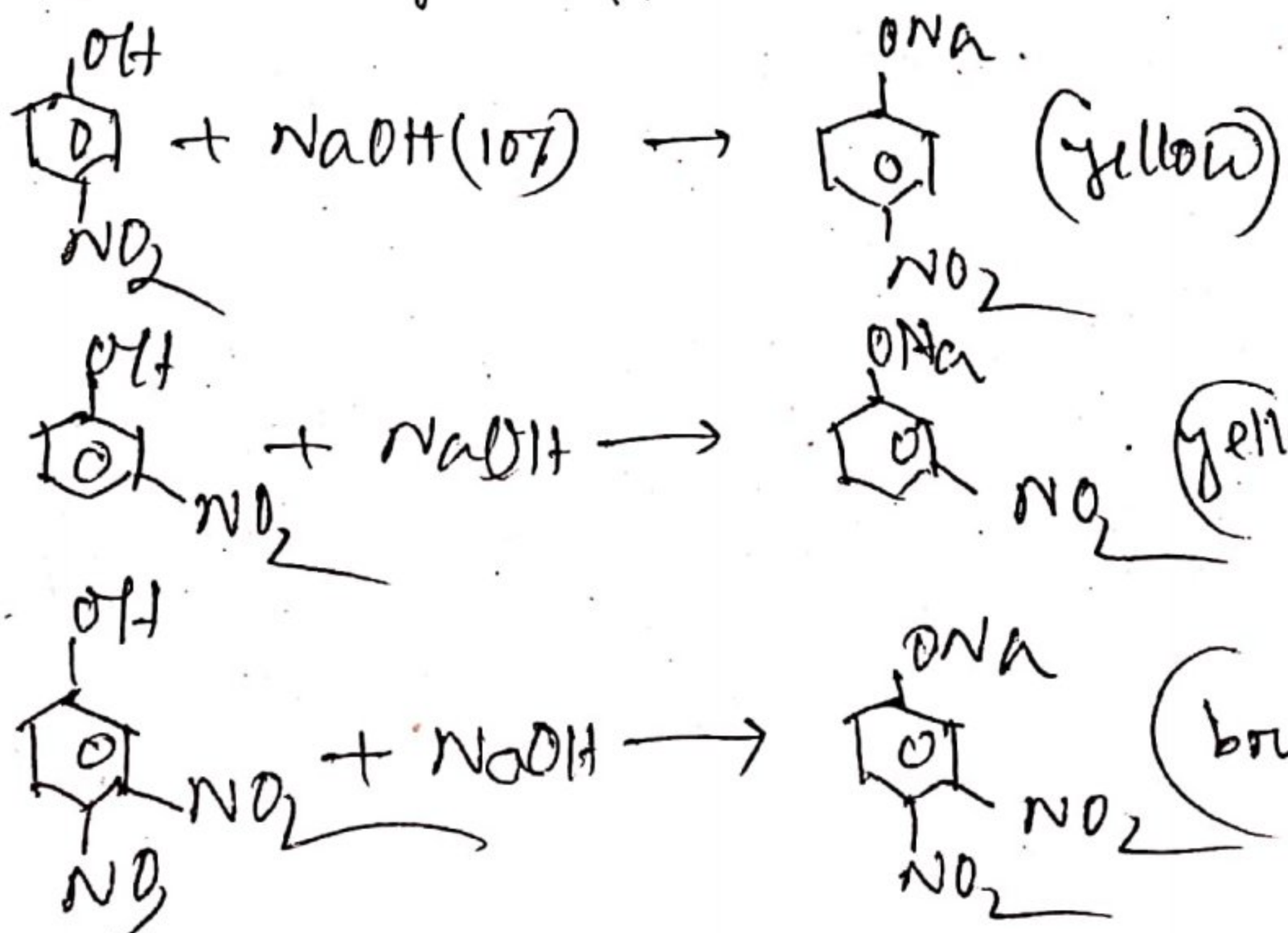
Mechanism:



⊞ Test for phenol



Test for Nitrophenol: violet



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⊞ Test for 1°, 2°, 3° alcohol

Lucas test: Conc HCl + $ZnCl_2$

$R_3OH \xrightarrow[\text{room temp}]{\text{conc. HCl + ZnCl}} R_3-Cl\downarrow + H_2O$

3°. immediately white ppt.

$R_2-CH-OH \xrightarrow[(1-5\ min)]{\text{conc. HCl + ZnCl}} R_2CH-Cl\downarrow + H_2O$

2° white ppt.

$R-CH_2-OH \xrightarrow[\text{long time}]{\text{Conc. HCl + ZnCl}} RCH_2-Cl\downarrow + H_2O$

1° white ppt

⊞ Test for estn:

$R-\overset{\overset{O}{\|}}{C}-OR' + H_2N-OH \xrightarrow[\text{Oxime}]{} R-\overset{\overset{P}{\|}}{C}-NH-OH + R'OH$

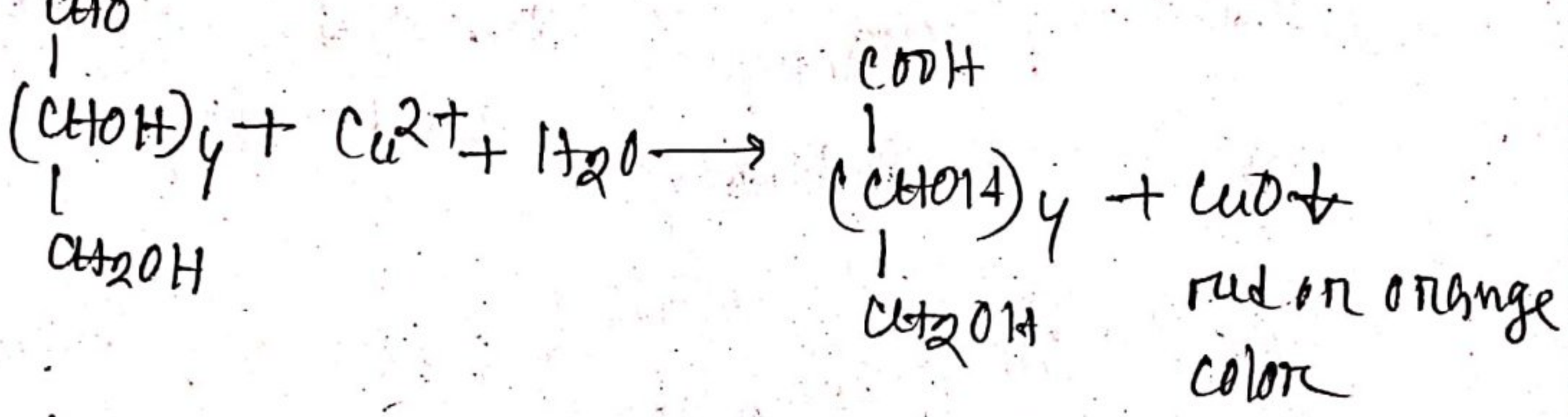
$R-\overset{\overset{O}{\|}}{C}-NH-OH + FeCl_3 \rightarrow \left(R-\overset{\overset{O}{\|}}{C}-NH-O\right)_3 Fe + HCl$

Bluish red or reddish brown. blue.

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Differentiating the monosaccharide from Disaccharide.

(1) Barfoed's Reagent test:



This indicate monosaccharide.

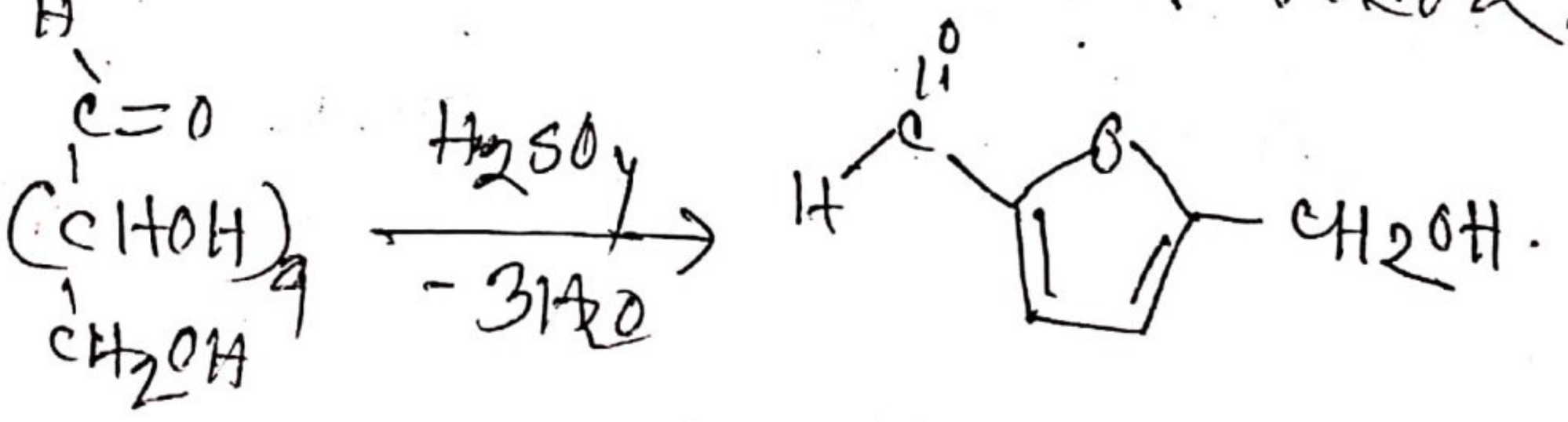
Carbohydrate test:

(1) Molish test:

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Distinguishing Ketoses from Aldoses:

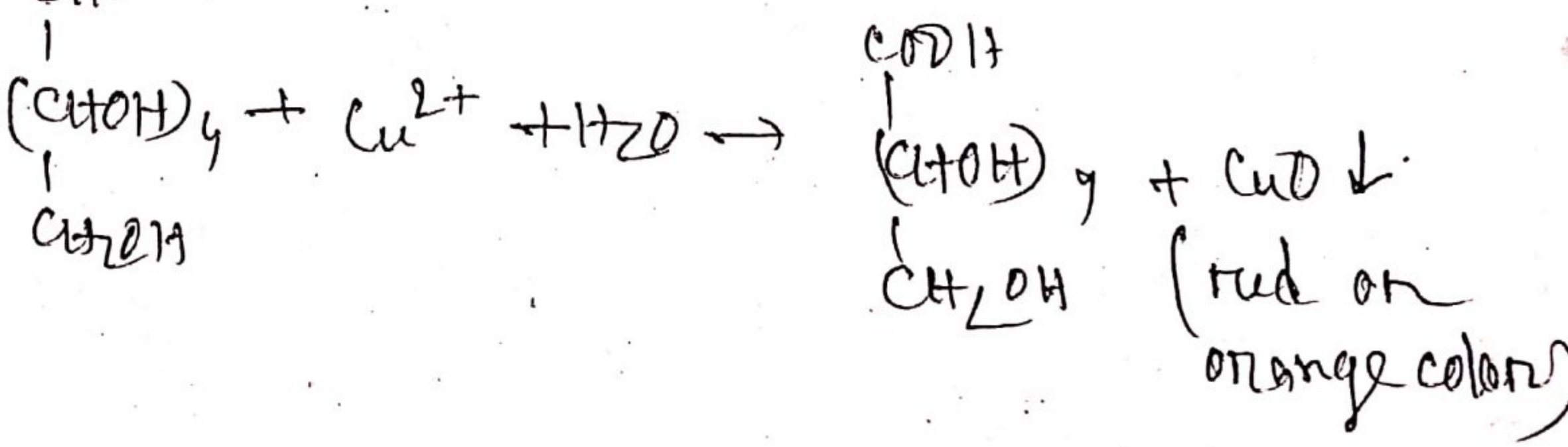
Seliwanoff's test: Seliwanoff's test is a chemical test which distinguishes between aldose and ketoses sugar. Ketoses are distinguished from aldoses via their ketone/ aldehyde functionality. If the sugar contains a ~~ketone~~ ketone group, it is a ketose. If a sugar contains a aldehyde group it is an aldoses. This test relies on the principle that, when heated ketoses are more rapidly develops its red color within 2 min on the other hand long standing or prolonged heating will developed the color with aldose.



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⊞ Distinguish mono and descharide

Barfoed's Reagent test: Monosaccharides are easily oxidized than the disaccharides By Barfoed's reagent A sample taking tube place in a bath of boiling water then remove from bath and cool. A yellow orange or orange-red precipitate is a positive test for monosaccharide If red precipitate is formed only after a ~~long~~ prolonged heating then the presence of disaccharide is indicated.



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Elementary test :

$$C, N, S, X \xrightarrow{Na;\ heat} NaCN, Na_2S, NaCNS, NaX.$$

Detection of sulphur :

1. Lead acetate test :

$$Na_2S + (CH_3COO)_2Pb \longrightarrow PbS\downarrow + CH_3COONa$$

(brown to black)

2. Sodium Nitro-prusside test :

$$Na_2S + Na\left[Fe(CN)_5NO\right] \longrightarrow Na_4\left[Fe(CN)_5NOS\right]$$

(Sodium Nitro-prusside)

Sodium thio-Nitro prusside.

→ purple colour

Detection of Nitrogen :

1. Prussian Blue test :

$$NaOH + FeSO_4 \xrightarrow{\Delta} Fe(OH)_2 + Na_2SO_4.$$

$$Fe(OH)_2 + 6NaCN \xrightarrow{\Delta} Na_4\left[Fe(CN)_6\right] + 2Na$$

Sodium ferro-cyanide (

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$$Na_4[Fe(CN)_6] + FeCl_3 \longrightarrow Fe_4[Fe(CN)_6]_3$$

Ferric ferro-cyanide

→ prussian blue

⊞ Combine test for N and S :-

$$NaCNS + FeCl_3 \longrightarrow Fe(CNS)Cl_2 + NaCl$$

Ferric-thio-cyano-chloride.

→ (Blood red)

⊞ Detection of halogens :

$$NaCN + HNO_3 \xrightarrow{\Delta} NaNO_3 + HCN\uparrow$$

$$Na_2S + HNO_3 \xrightarrow{\Delta} NaNO_3 + H_2S\uparrow$$

Blue test :

$$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$$

(white)

$$NaBr + AgNO_3 \longrightarrow AgBr + NaNO_3$$

(pale yellow)

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$$NaI + AgNO_3 \longrightarrow AgI + NaNO_3$$
(yellow)

Difference of Cl, Br, I.

(i) Reaction with: Nitric acid :

$$NaCl + HNO_3 \xrightarrow{CCl_4} NaNO_3 + HCl.$$

$$HCl + HNO_3 \longrightarrow Cl_2 \text{ (in } CCl_4 \text{ colourless)}$$

$$NaBr + HNO_3 \xrightarrow{CCl_4} NaNO_3 + HBr$$

$$NBr + HNO_3 \longrightarrow Br_2 \text{ (in } CCl_4 \text{ deep brown)}$$

$$NaI + HNO_3 \xrightarrow{CCl_4} NaNO_3 + HI$$

$$HI + HNO_3 \longrightarrow I_2 \text{ (in } CCl_4 \text{ violet)}$$

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Functional Group test

Amine test :

Test for primary aliphatic amine :

1. Nitrous acid test :

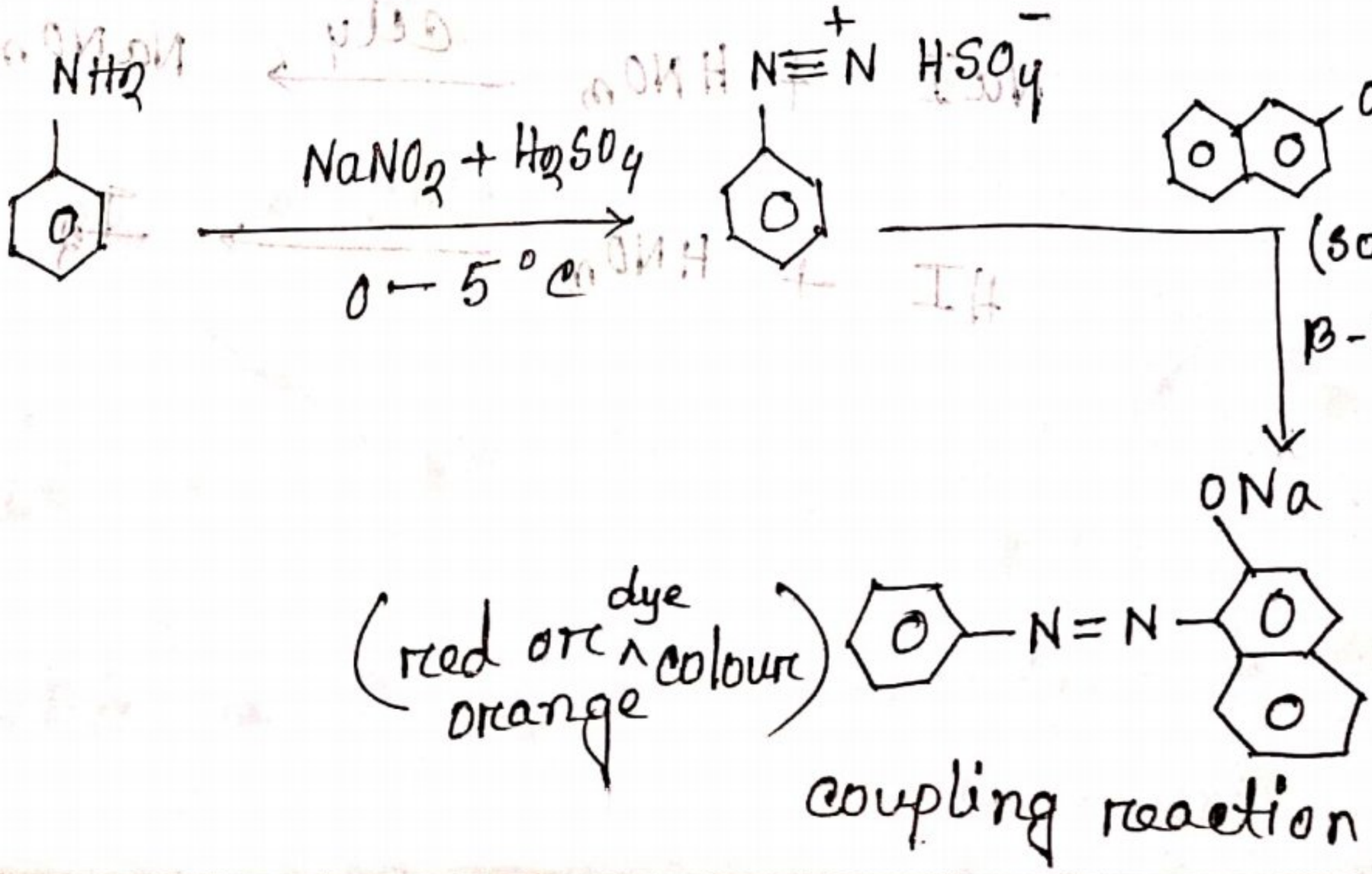
$$10\%\ NaNO_2 + HCl \xrightarrow{0-5^\circ C} HNO_2$$

$$HNO_2 + R-NH_2 \longrightarrow R-\underset{\underset{H}{|}}{N}-N=O$$

(oily dark colour liquid)

Test for primary aromatic amine :-

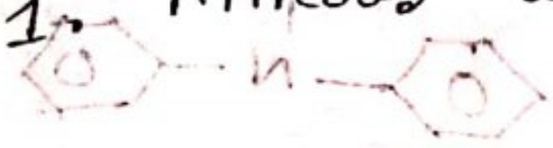
1. Diazotization test :



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Test for 2° amine :

1. Nitrous acid test :



$$R_2NH + HNO_2 \longrightarrow R_2N-N=O + H_2O$$

oily dark colour .



Nitroso reaction

2. Libermann's test :

$$HCl + NaNO_2 \longrightarrow HNO_2 + NaCl$$

$$R_2NH_2 + HNO_2 \longrightarrow R_2-N-N=O + H_2O .$$



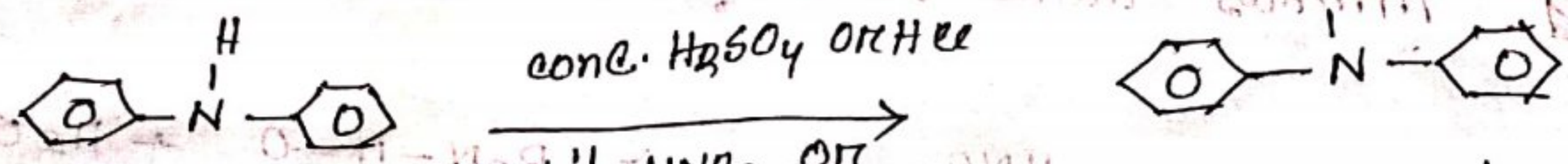
$$R_2-N-N=O \xrightarrow[\text{Conc. } H_2SO_4]{} \text{Blue coloration}$$

$\downarrow$ NaOH

(blue black colour)

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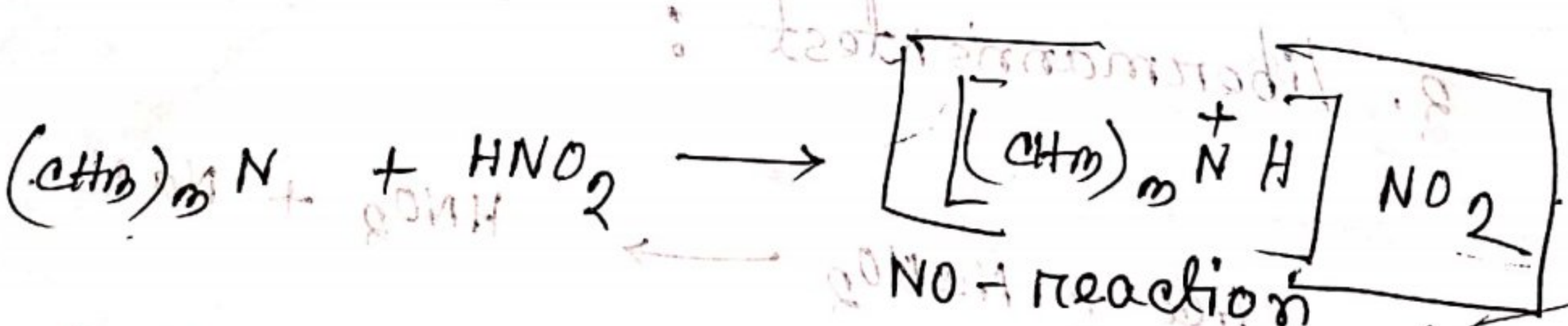
⊞ Specific test for di-phenyl amine :



→ deep blue colour

⊞ Tertiary Amines :

1. Nitrous acid test :



→ पीला रंग colourless द्रव



→ green and brown color.

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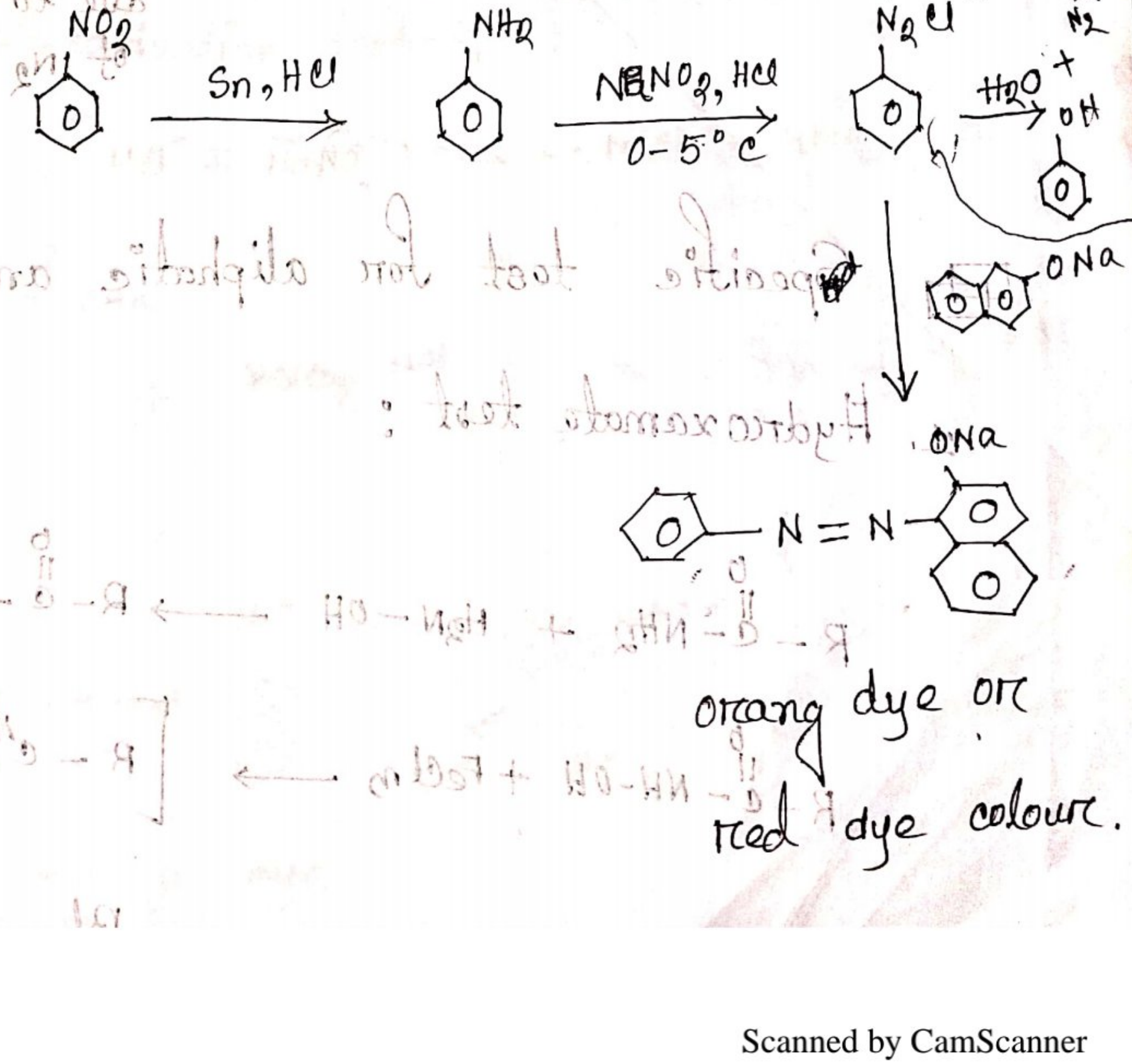
Nitro-compound test :

1. Griess reagent :

→ 1·0% Sulphanilic acid 

0·10% 1-naphthyl amine

2. Reduction of Nitro-compound :



orang dye or red dye colour.

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Special test for: di- and trinitro benzene.

Test for amide (General):

1. Nitrous acid test:

$$R-CO-NH_2 + HNO_2 \longrightarrow R-\overset{O}{\overset{\|}{C}}-OH + N_2\uparrow + H_2O$$

Brisk effervescence due to evolution of N_2 gas.

Specific test for aliphatic amide:

1. Hydroxamate test:

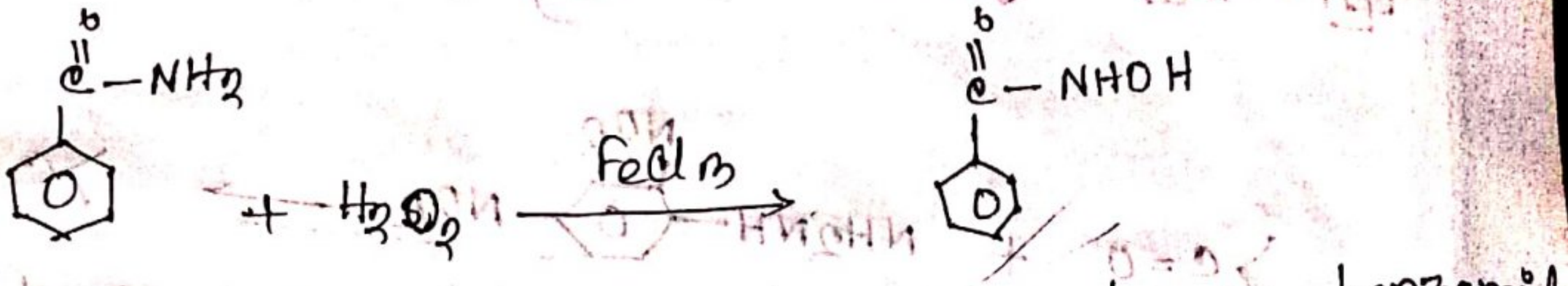
$$R-\overset{O}{\overset{\|}{C}}-NH_2 + H_2N-OH \longrightarrow R-\overset{O}{\overset{\|}{C}}-NH-OH + NH_3$$

$$R-\overset{O}{\overset{\|}{C}}-NH-OH + FeCl_3 \longrightarrow \left[R-C(=O)-N(H)-O^{(-)}\right]_3 Fe$$

Blue-red colour.

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Specific test for aromatic Amide

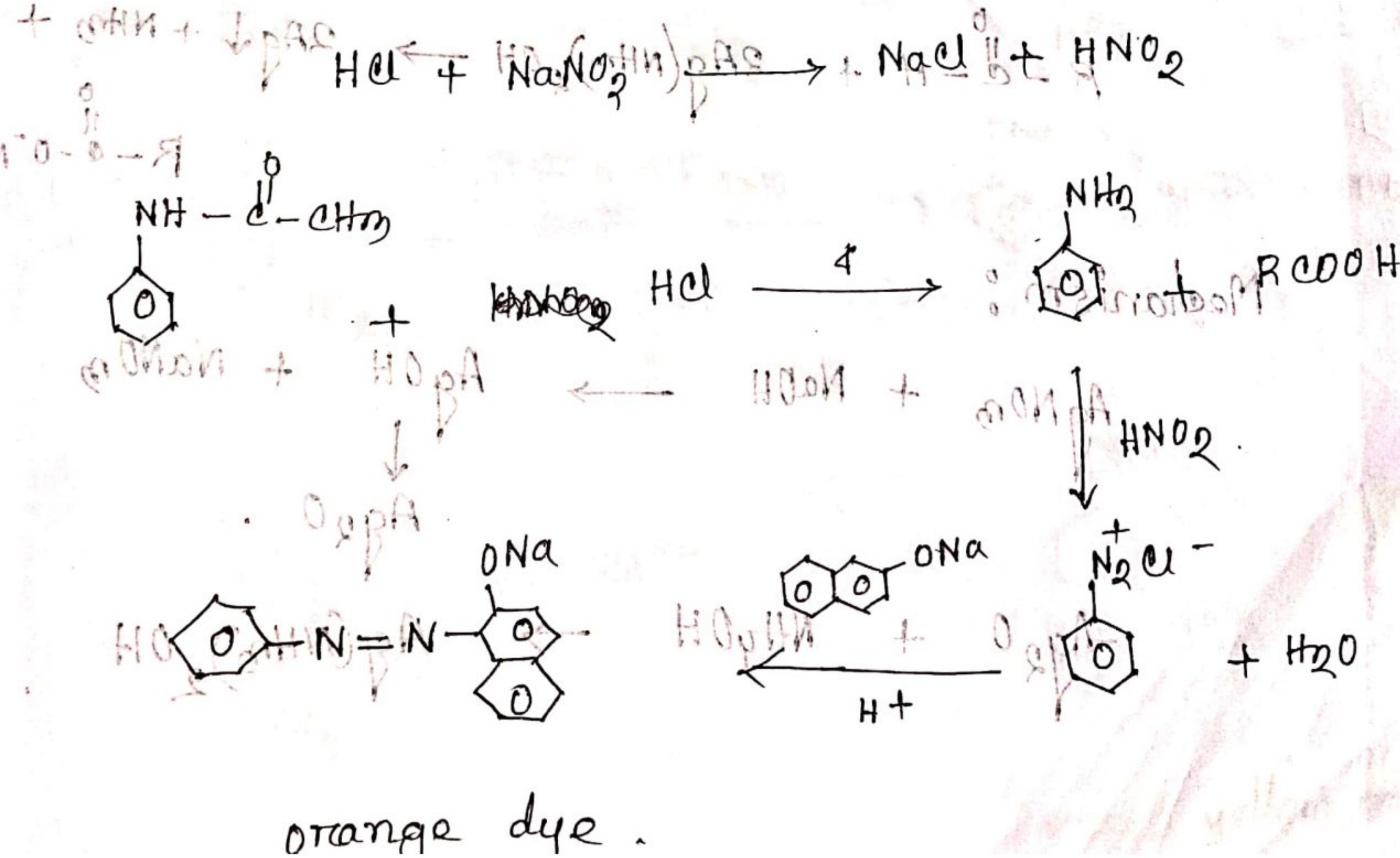


N-Hydroxy-benzamide

bluish red.

Test for anilide :

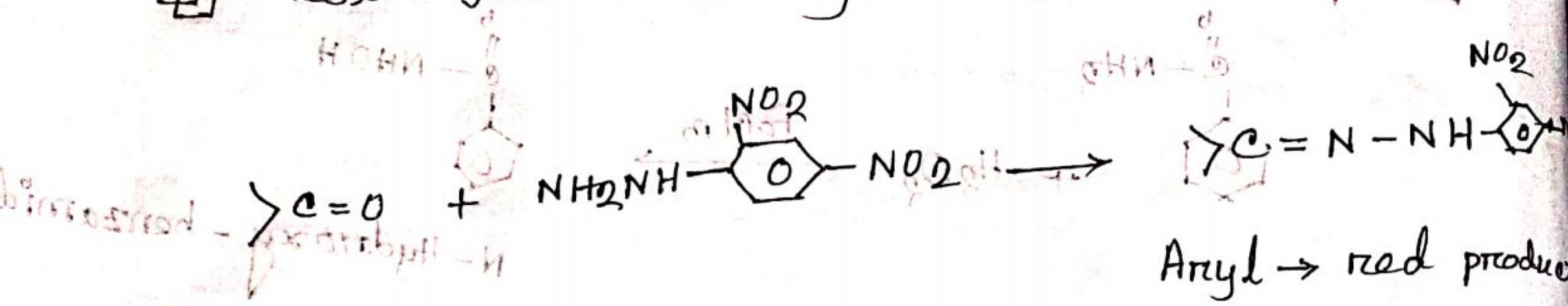
1. Diazotization test :



orange dye.

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Test for Carbonyl compound :



Aryl → red product

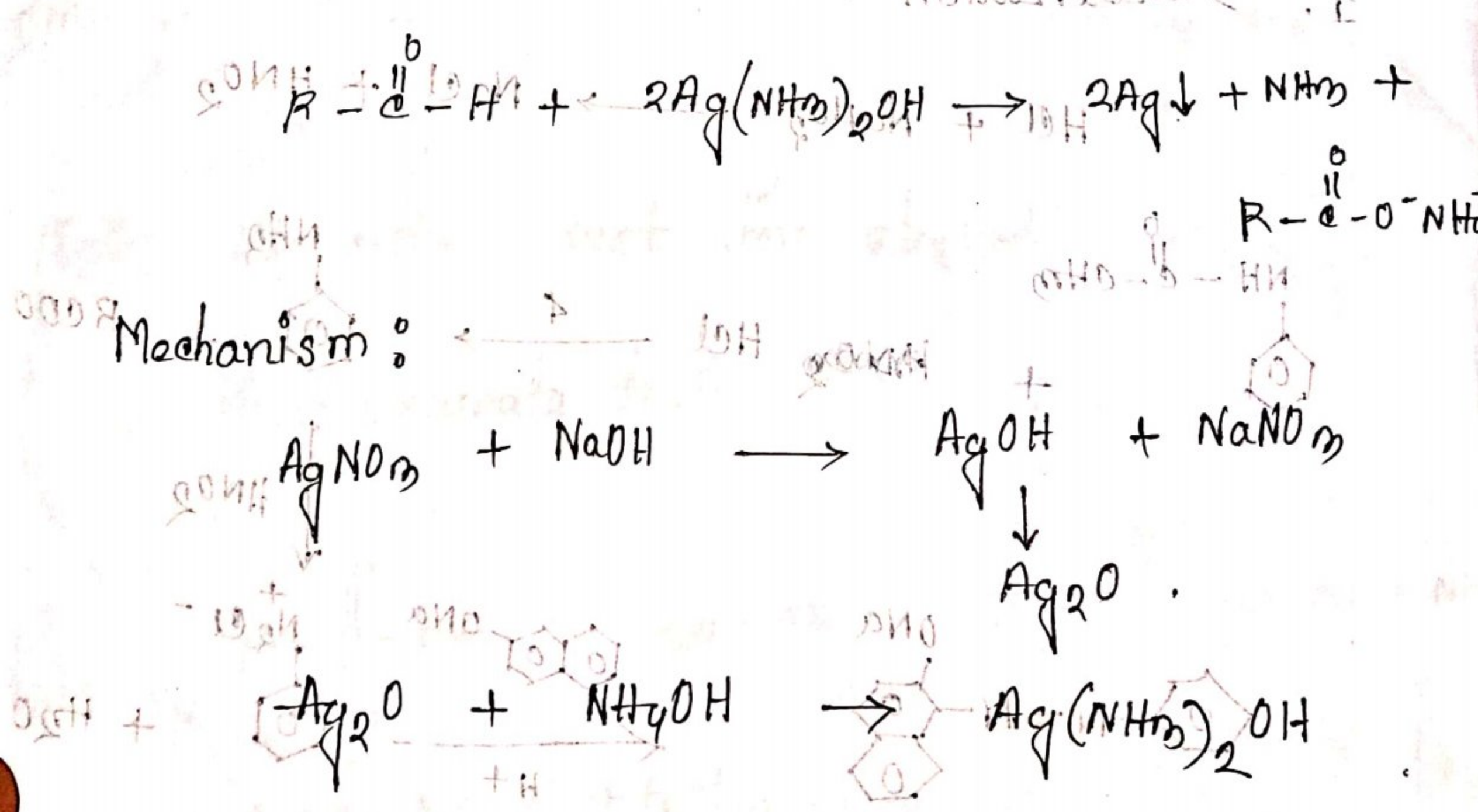
Aliphatic → Yellow -

Test for aldehyde :

1. Tollen's reagent for oxidation test :



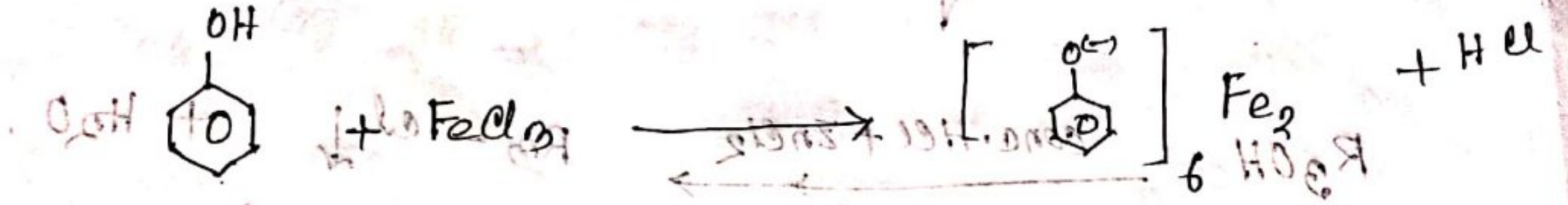
Mechanism :



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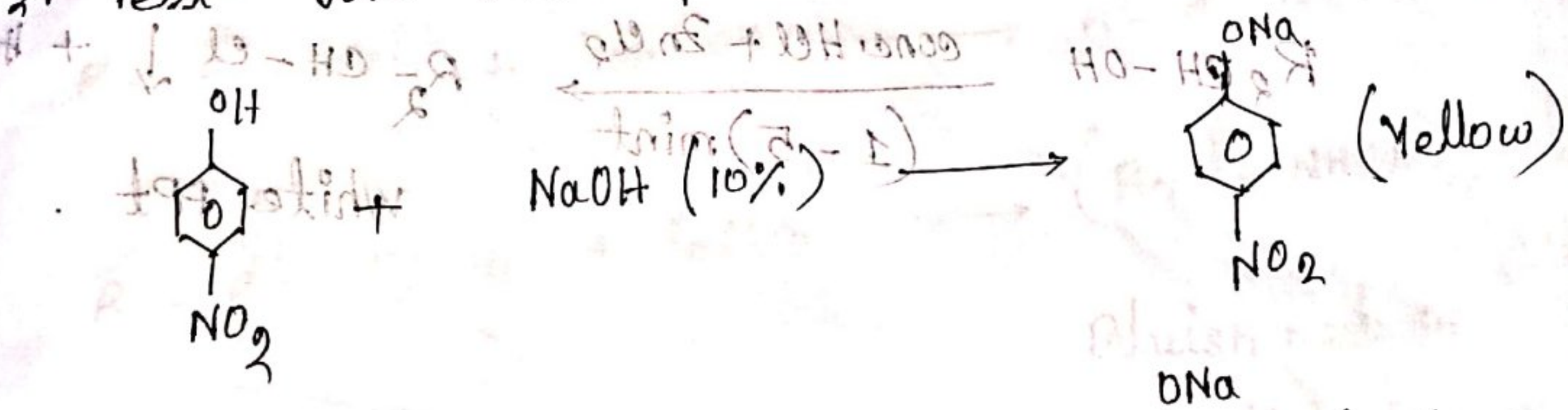
Test for phenol :-

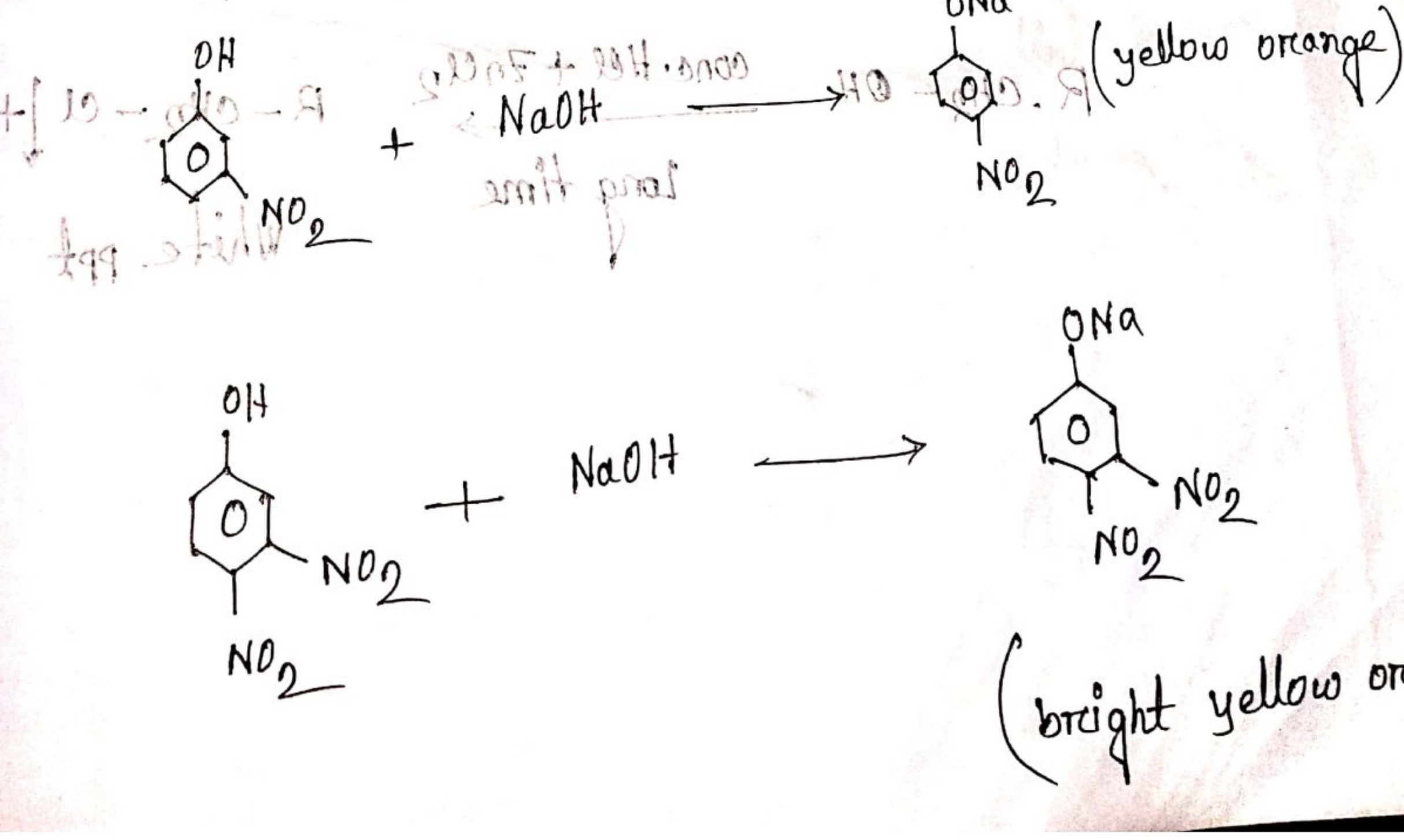
1. Ferric chloride test :



(Violet)

2. Test for Nitro-phenol :-





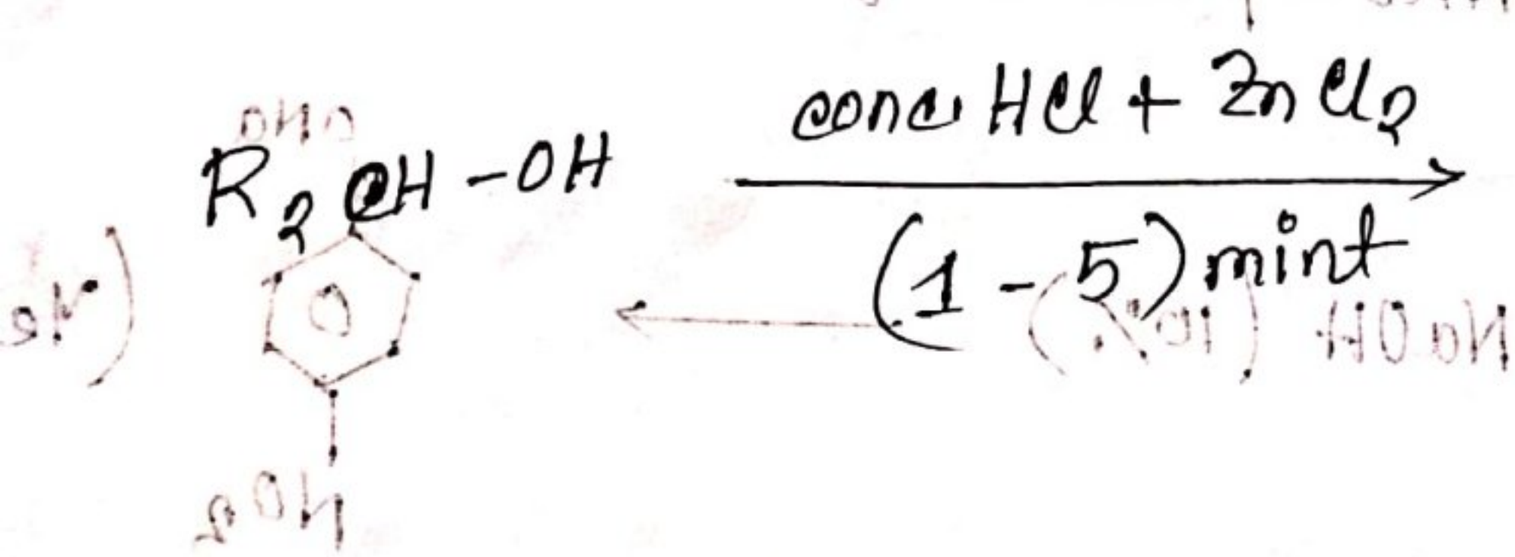
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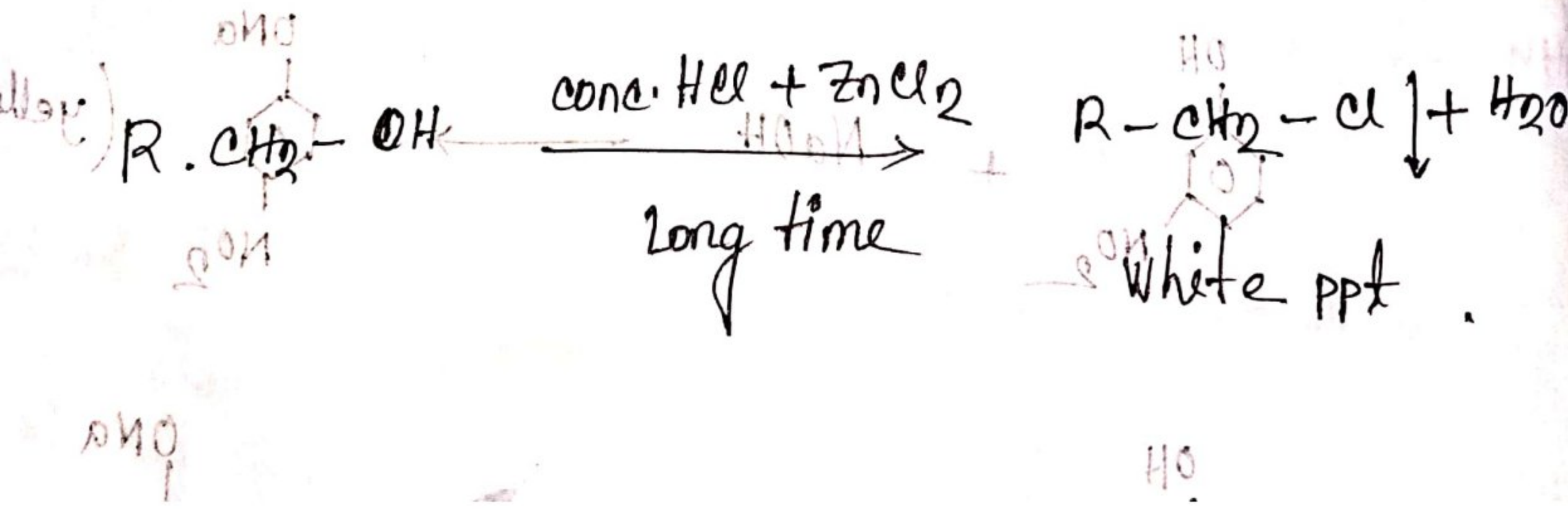
Test for 1° 2° 3° alcohol :

1. Lucus reagent test :



3° imidiately





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⊞ Test for Carboxylic Acid :

$$R-COOH + NaHCO_3 \longrightarrow RCOONa + CO_2\downarrow + H_2O$$

CO_2 - এর বুদ বুদ বের হবে ।

⊞ Test for ester :

$$R-\overset{O}{\overset{\|}{C}}-OR' + H_2N-OH \longrightarrow R-\overset{O}{\overset{\|}{C}}-NHOH + R'OH$$

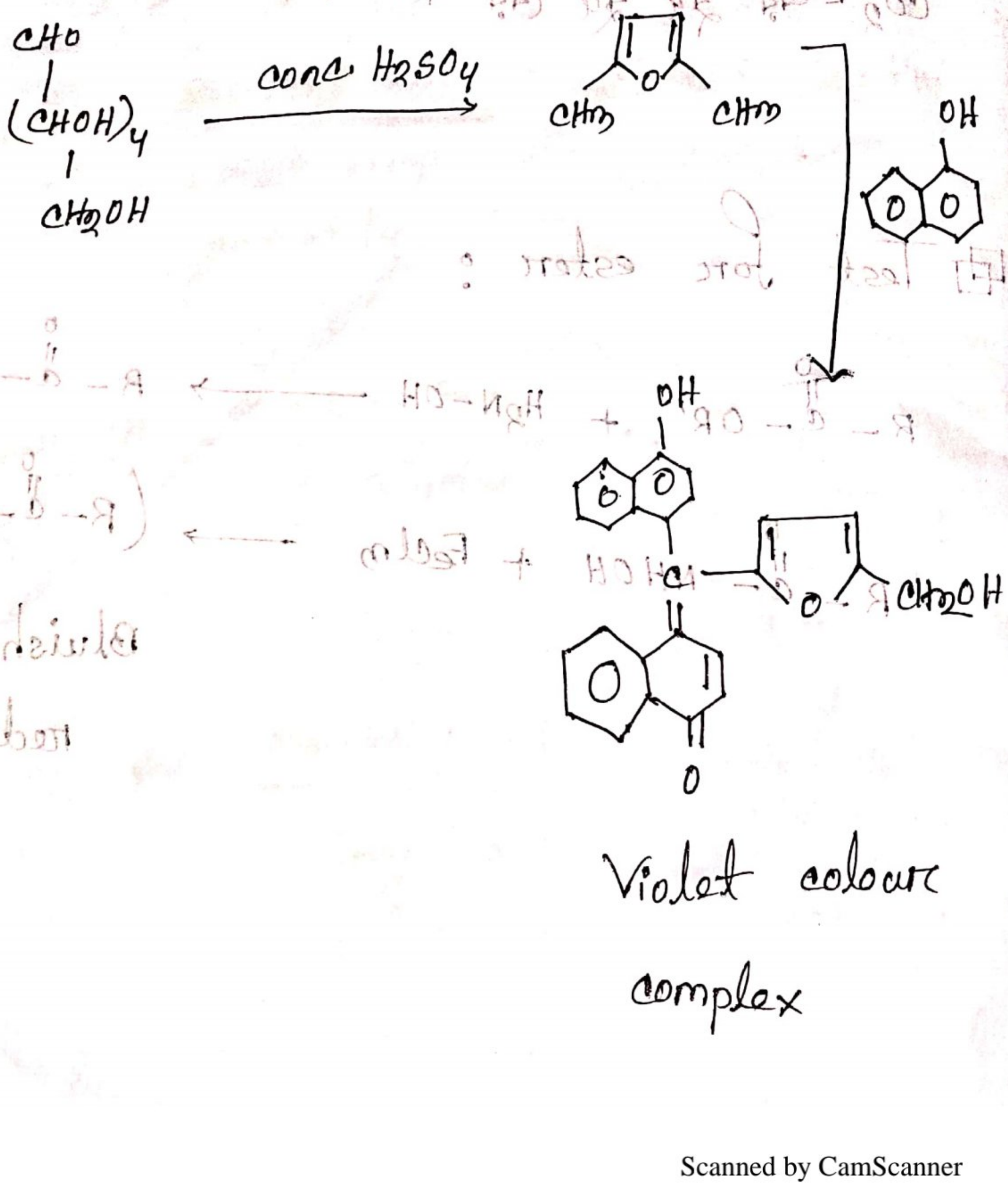
$$\left(R-\overset{O}{\overset{\|}{C}}-NHOH\right)_3 Fe + HCl$$

Bluish red or reddish brown .

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Carbohydrate test :

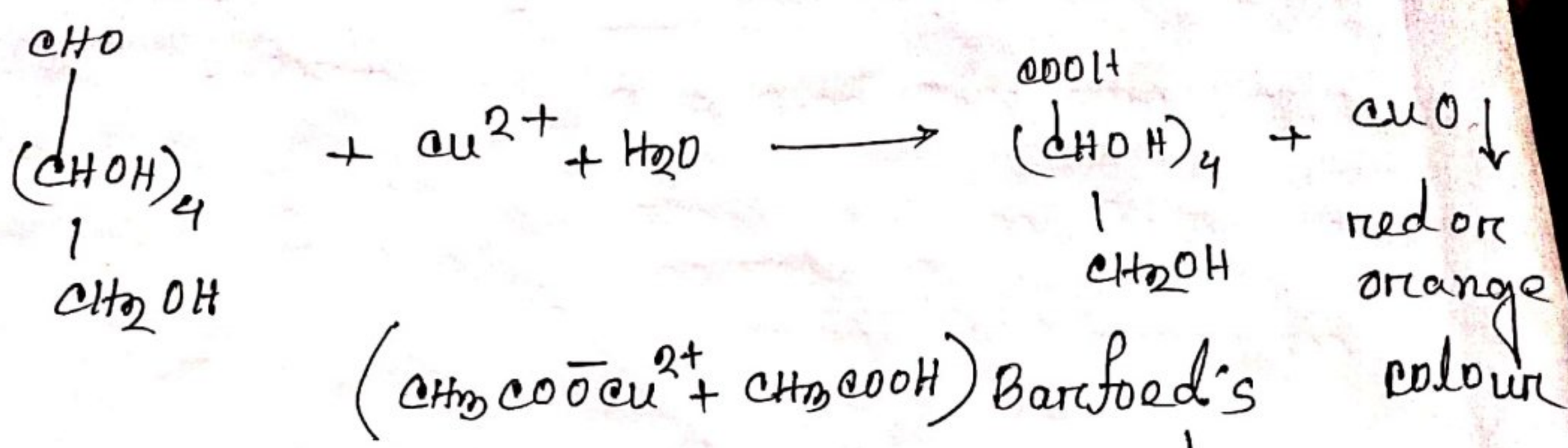
1. Molish test :



Violet colour complex

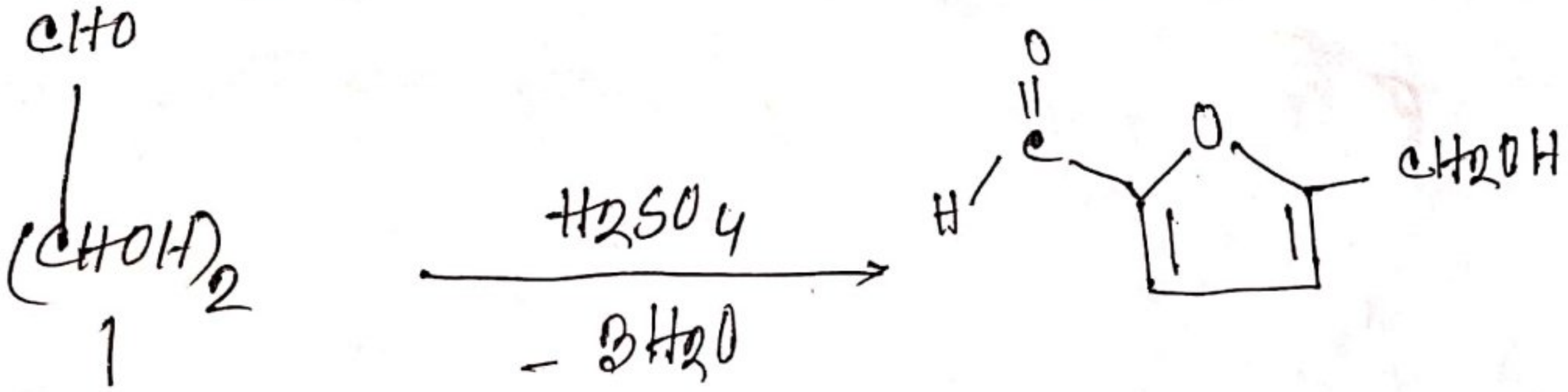
Differentiating the Mono-saccaride for Di-saccharide:

1. Barfoed's reagent test :



Distinguishing ketose from Aldose.

1. Seliwanoff's test :



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CH_3COOH দিয়ে acedifying করা হয় কেন? sulphar detection -এ

$$NaOH + CH_3COOH \longrightarrow CH_3COONa + H_2O$$

যদি direct lead acedate দিলে,

$$NaOH + (CH_3COO)_2Pb \longrightarrow Pb(OH)_2 \text{ black ppt.}$$

$$Na_2S + (CH_3COO)_2Pb \longrightarrow PbS \downarrow \text{ black ppt.}$$

Same ppt পাচ্ছ তাই S identifying করা কঠিন হয়ে যায়।

Sodium Nitro-prussiide test :

$$Na_2[Fe(CN)_5NOS] + Na_2S \longrightarrow Na_4[Fe(CN)_5NOS]$$

Ferric ferro-cyanide.

$$Fe_4[Fe(CN)_6]_3 \rightarrow \text{Prussian blue.}$$

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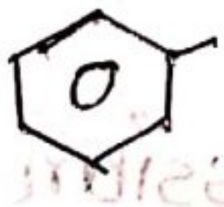
Q. Why we take mixed melting point ?

→ pure compound এবং যে compound টি পাওয়া গেছে তা একসাথে mix করে তার m.p এবং b.p নিলে যেই m.p/b.p আসে pure compound এর m.p or b.p সমান হয় তবে বুঝা যাবে compound টি pure.

Q. Why we use $AgNO_3$?

→ Because organic compound.

Q.  + $AgNO_3$ ⟶ No-reaction why ?

→ Because  is a covalent compound and $AgNO_3$ is an ionic compound.

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Q. To make - an covalent ionic compounds from covalent compound what kind of metal are used ??

→ Highly electropositive metal (Na)

Q. What is the necessity of sodium fusion ?

→ To make ionic compound.

Q. In sodium fusion method why we use MeOH.

→ To form sodium methoxide from unreative-Na.

Q. Why use do not use normal organic solvent such as CCl_4, $CHCl_3$, CH_2Cl_2, CH_3OH in this test ?

→ By Because by using this we can not classify the compared.

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Sulphate - detection

Q. Why do we need to acidify the solution with dilute acetic acid?

→ During the fusion process if any trace of sodium metal remains unreacted, it will from NaOH with water in that case NaOH may react with $Pb(CH_3COO)_2$ to give black ppt of $Pb(OH)_2$ which may hesitate us.

$$NaOH + CH_3COOH \longrightarrow CH_3COONa + H_2O$$

So, we need to add dil. CH_3COOH to neutralize the NaOH if present.

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Na. fusion:

1. Why sodium metal is used in lassaigne test?

→ Na metal is very reactive, that's covalently bonded N, S, X etc in organic compounds can form ionic compounds by reacting with Na. and produced ionic sodium salts. are soluble in water.

Q. What would happens when sodium be deficient in fusion test?

→ It is essential to use an excess of sodium otherwise if sulpher and nitrogen are both present sodium thiocyanate ($NaSCN$) may be produced in the test for Nitrogen it may give a red coloration with ferric ions but no prussian blue.

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Since there will be no free cyanide ions with excess of sodium the thiocyanate will be decomposed :-

$$NaSCN + 2Na \longrightarrow NaCN + Na_2S.$$

Q. Why methanol is used in Lassaigne test?

→ After heating the composition of sodium and sample, it is cooled to room temp. Then we add 5-7 drops of methanol before adding water. If the composition after being cooled contains any sodium, it is removed as $CH_3O^-Na^+$ by adding CH_3OH. Otherwise if we don't add CH_3OH before using water, Na will react vigorously with H_2O to result damage

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Nitrogen detection :

Q. Why KF is added in N identification?

→ The addition of the minute amount of KF facilitates the conversion of extra Fe^{3+} ion to $K_3[FeF_6]$ (Potassium ferrifluoride) which leads to the formation of a pure prussian blue.

$$KF + Fe^{3+} \longrightarrow K_3[FeF_6]$$

Q. Why HCl was not used in acidifying the alkaline solutions?

→ We usually use dil. H_2SO_4 but HCl must be avoided otherwise $FeCl_3$ can

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formed which is of yellow colour and can cause the prussian blue to appear greenish.

Q Why H_3PO_4 is used when after the addition of dil. H_2SO_4 the colour of $Fe_4[Fe(CN)_6]_3$ appears greenish?

⟶ This existance of greenish colour usually occurs due to the presence of persisting Fe^{3+} ion which is then immediately converted to a colourless water soluble complex $[Fe(HPO_4)^+ aq]$. Then the prussian blue appears if N is present.

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Halogen Detection

Q. Why cyanide or sulphide ions must be removed before adding silver ions ? to test for the halides ?

→ The Lassaigne sample, when selected to test for halide ions, it is abrious to remove cyanide and sulphide ions, if any in sample -

$$AgNO_3 + CN^- \longrightarrow \underset{\text{ppt}}{AgCN\downarrow} + NO_3^-$$

$$AgNO_3 + S^{2-} \longrightarrow \underset{\text{ppt}}{Ag_2S\downarrow} + NO_3^-$$

We see that if CN^- and S^{2-} ions are not removed, they will react with Ag^+ to result in positive hallide test though the sample does not really contain any halogen.

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Q. How to remove CN^- and S^{2-} ions ?

→ We can remove CN^- and S^{2-} ions from the sample by adding conc. HNO_3 acid to it.

$$NaCN + HNO_3 \xrightarrow{\Delta} NaNO_3 + HCN\uparrow$$

$$Na_2S + HNO_3 \xrightarrow{\Delta} NaNO_3 + H_2S\uparrow$$

$NaNO_3$ being a soluble salt does not interfere further. [HCN উড়ে চলে যায়]

→ HNO_3 ছাড়া অন্য কোন Acid use করা যাবে না তাহলে ppt পড়বে। H_2SO_4 use করলে Na_2SO_4 এর ppt ও পড়বে।

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Q. Why the sample after being acidified with conc. HNO_3 need to boiled ?

→ The removal of HCN/H_2S made from the reaction of CN^- and S^{2-} ions with HNO_3 is effected by boiling the fusion solution to reduce it to half of its initial volume.

The sample after being acidified with conc. HNO_3 to remove HCN and H_2S to boil.

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→ The characteristic reaction of carbonyl group is nucleophilic addition.

→ The characteristic reaction of benzene is Nucleophilic ~~add~~ substitution and electrophilic addition.

→ Tollen's reagent $[Ag(NH_3)_2OH]$ → oxidizing agent

$-\overset{O}{\overset{\|}{C}}-H$ oxidize করে $\longrightarrow -COOH$.

যেহেতু $Ag(NH_3)_2OH$ mild oxidizing agent তাই—

ketone কে oxidized করতে পারে না।

→ বেশি Heat দিলে $Ag(NH_3)_2OH$ বিয়োজিত হয়ে

$Ag\downarrow$ পড়বে। ফলে reaction-এর মাধ্যমে $Ag\downarrow$ পড়বে না।

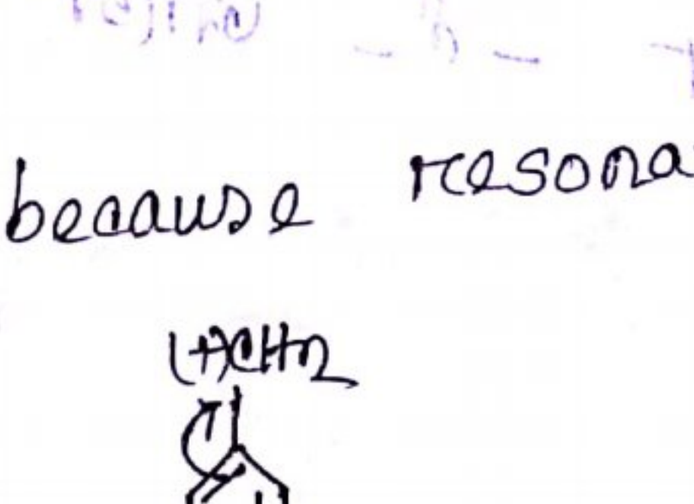
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Functional Group

⊞ Test for alcohol :

→ Distinguishing between primary, secondary and tertiary alcohol by Lucas reagent ($ZnCl_2$ + conc. HCl)

→ Aliphatic 3° alcohol এর reaction করে 3° carbocation form করে, it is more stable than other carbocation and এটা Hyperconjugation. → $ZnCl_2$ carbon oxygen bond কে weak করার জন্য use করা হয়।



→ Aromatic 1° alcohol more stable than 3° alcohol because resonance stabilised.



⊞ Test for Carbohydrates

→ Distinguishing between mono-saccharides from di-saccharides by Barfoed's reagent (($(CH_3COO)_2Cu$ + acetic acid))

→ Mono-saccaride first reaction দেয়। but di-saccaride slowly দিবে বা প্রথমে free functional group থাকে না। ভাঙ্গার পর free functional group হয়। তাই দেরিতে ppt পড়ে।

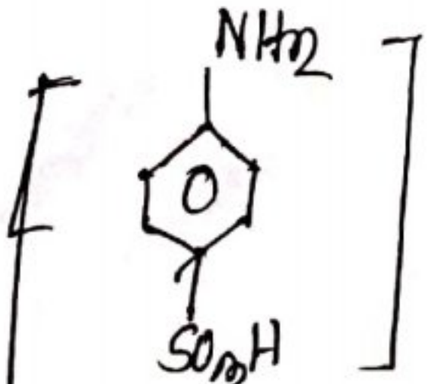
→ Sucrose - এ $-\overset{O}{\overset{\|}{C}}-$ এর ppt পড়ে বা $-\overset{O}{\overset{\|}{C}}-$ জাতীয় reaction দেয়। $-CHO$ এর মিরর

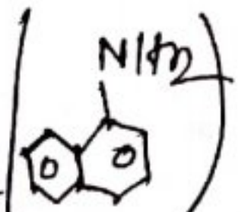
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→ Distinguishing between aldose and ketose by Selliwanoff's test (A solution of resorcinol and HCl or conc. H_2SO_4).

Test for Nitro-compound,

→ Griess reagent (1.0% sulphanilic acid solution



and 0.10% 1-napthyl amine 

⇒ Spectroscopic Analysis Liquid sample

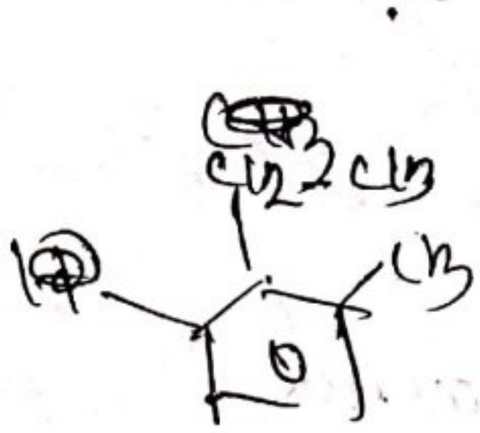
-এর জন্যে ~~পরে~~ প্রযোজ্য।

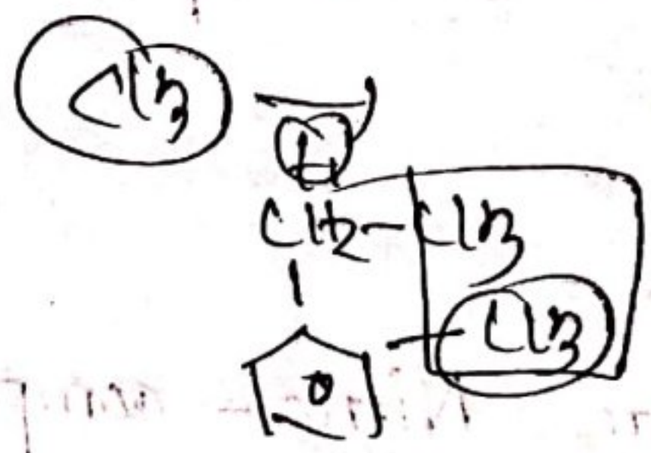
→ Amino-compound থাকলেও oxidized হতে পারে তাই NO_2-group test confirm দেয়।

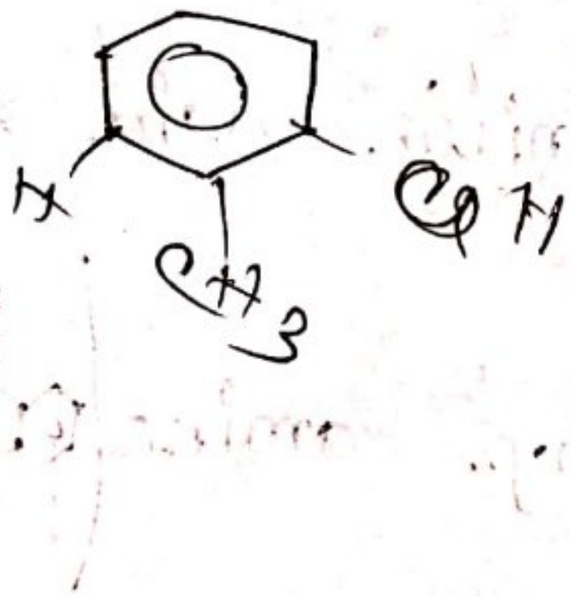
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Q. Organic compound colour হয় কেন?

→ Conjugation-এর জন্য।





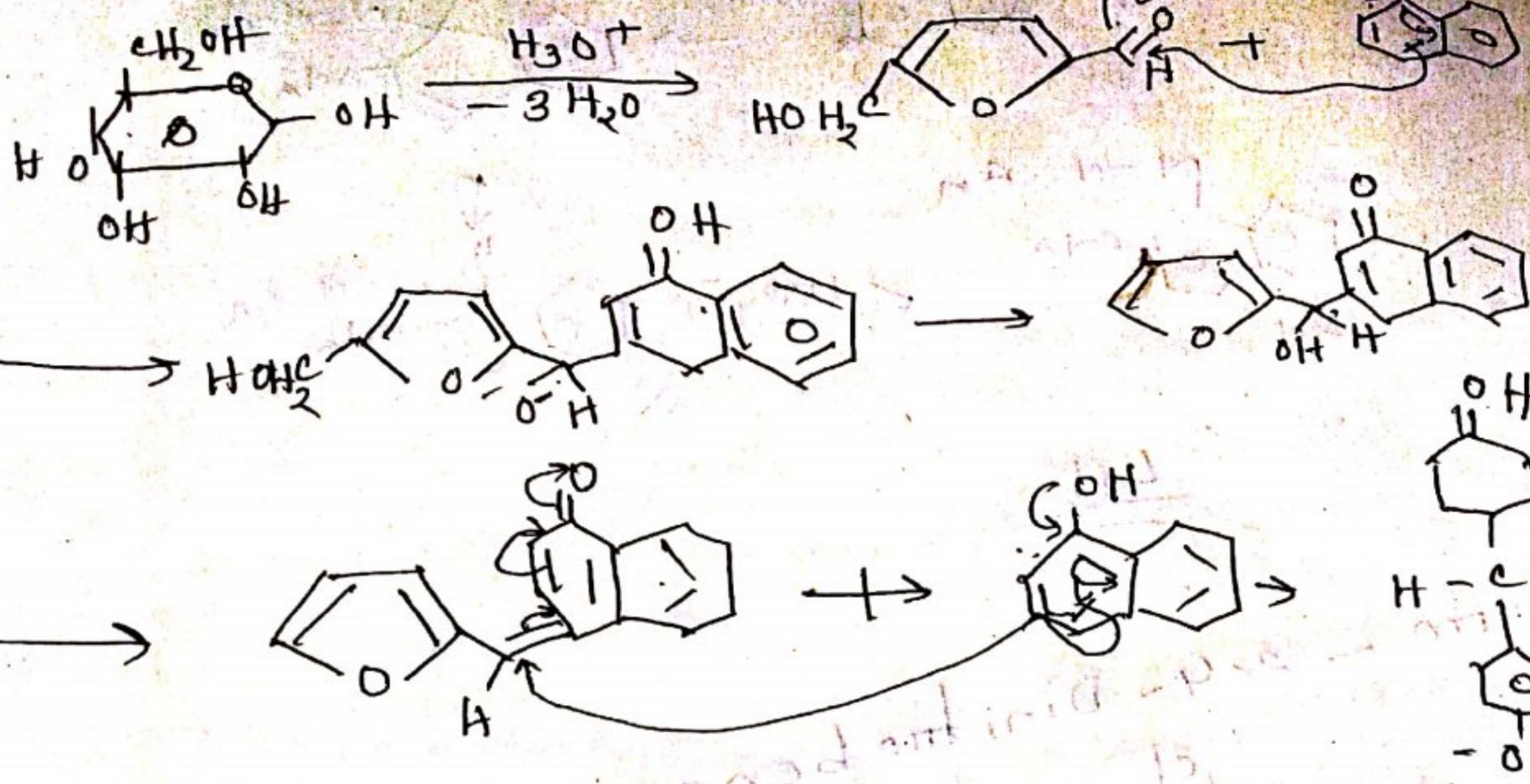






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carbohydrate test: mechanism:



Diazotization mechanism:

1) $HONO + H_3O^+ + HSO_4^- \longrightarrow H_2O-NO + H_2O$

$H_2O-NO \rightarrow NO^+$

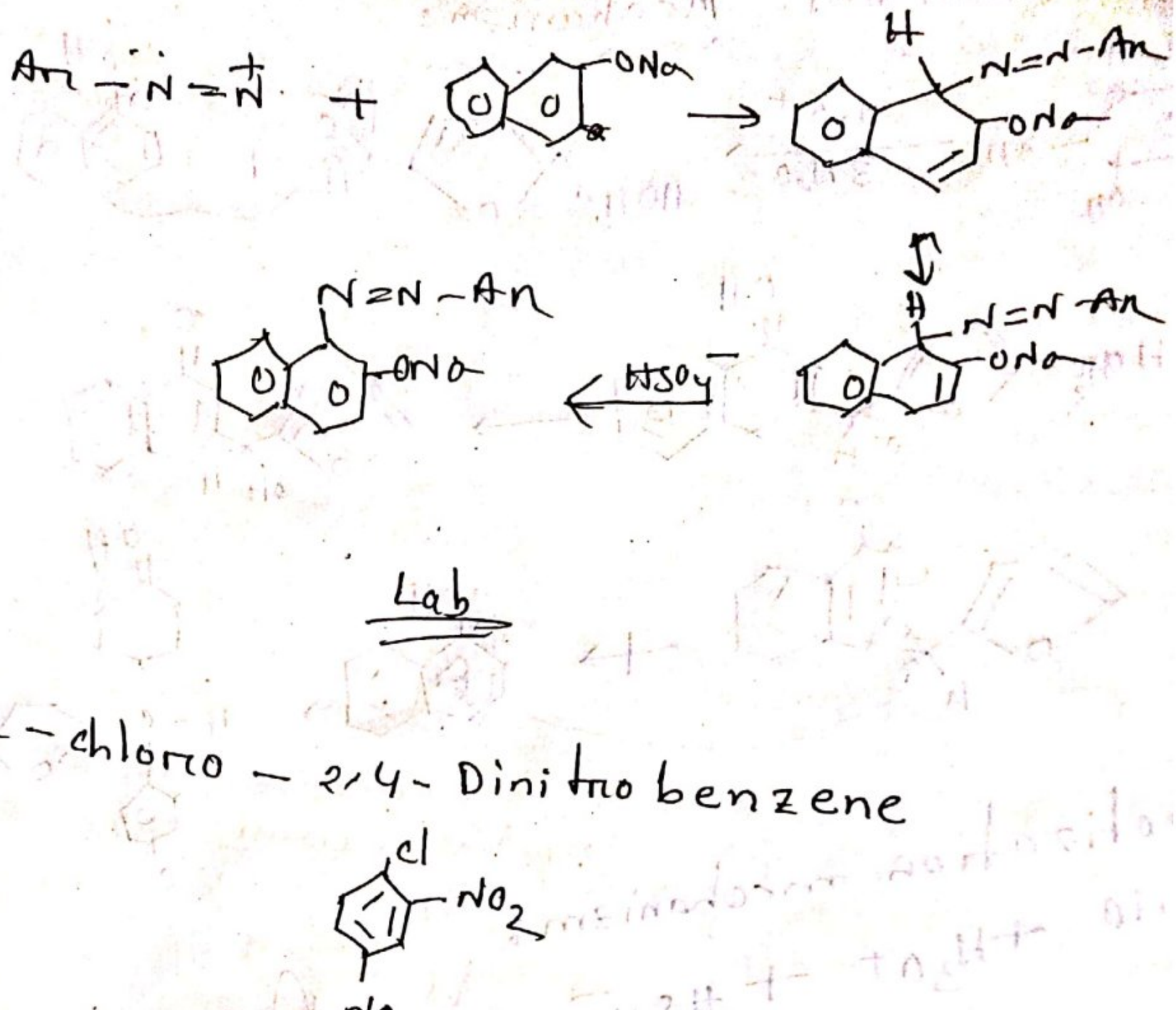
2) $Ar-\ddot{N}H_2 + NO^+ \longrightarrow Ar-\overset{+}{N}H_2-N=O$

$\longrightarrow Ar-N=N-OH \longrightarrow Ar-N=N-OH_2$

$\xrightarrow{-H_2O} Ar-N\equiv N \leftrightarrow Ar-\ddot{N}=\overset{+}{N}$

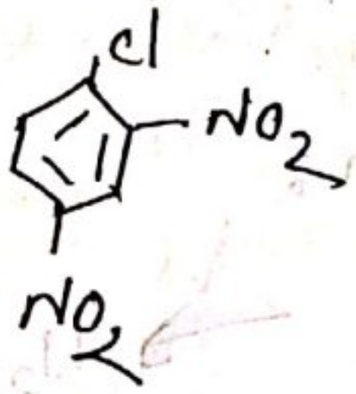
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4)

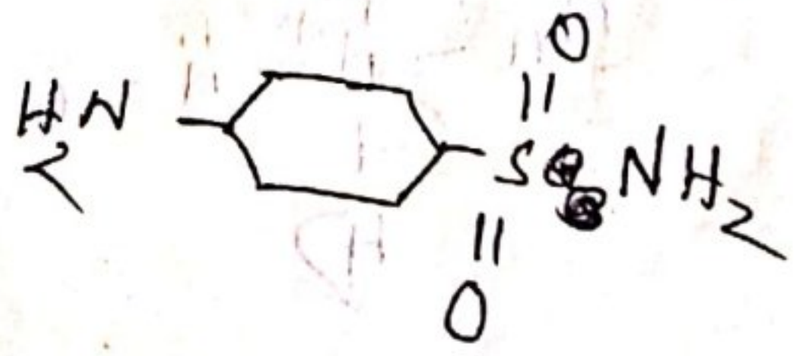


Lab

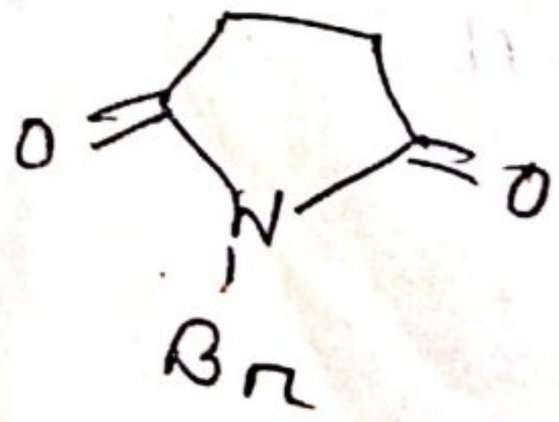
* 1-chloro-2,4-Dinitrobenzene



* Sulphanilamide:

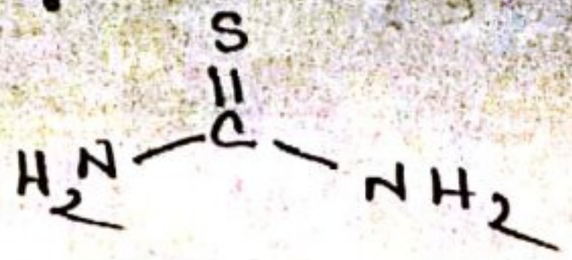


* N-Bromosuccinamide:

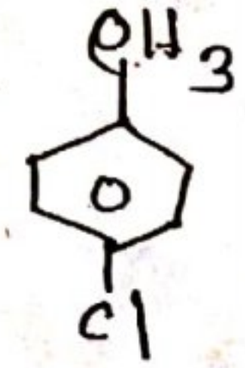


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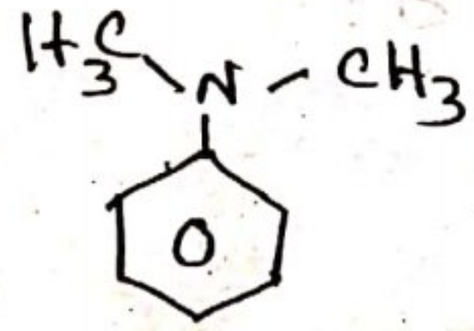
* Thioarea :



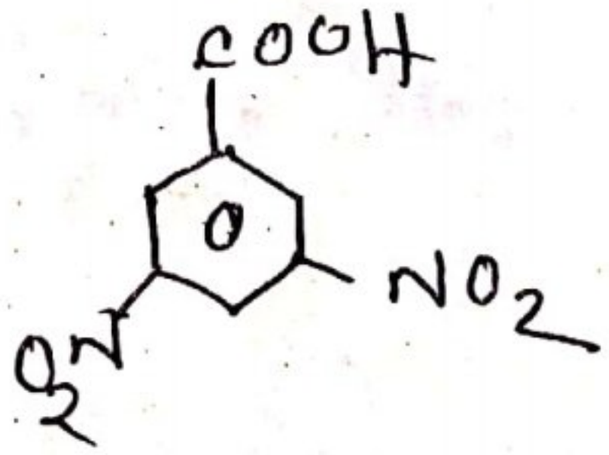
* 4-chlorotoluene :



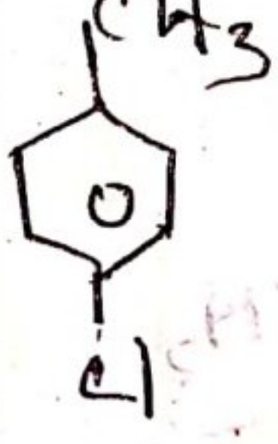
* NN-Dimethylaniline :



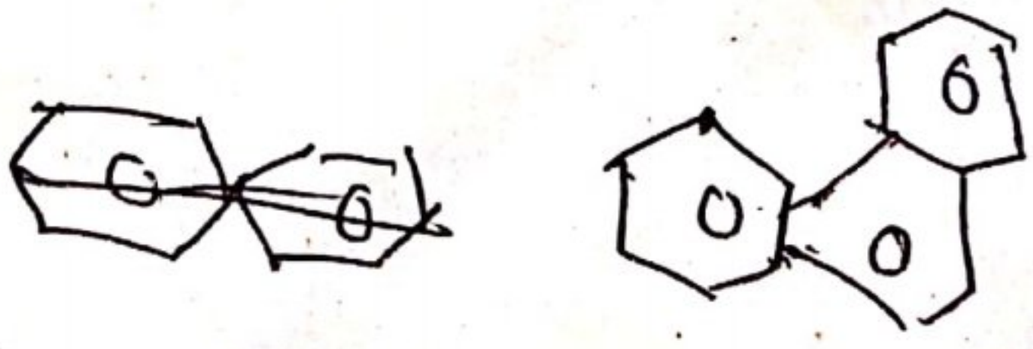
* 3,5 - dinitrobenzoic acid :



* P-chloroaniline

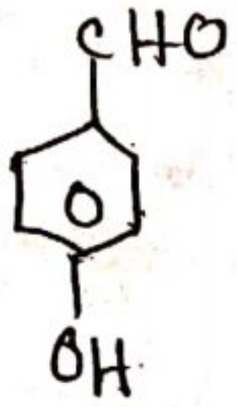


* Anthracene :

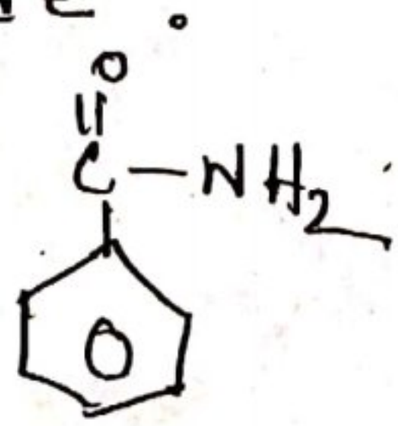


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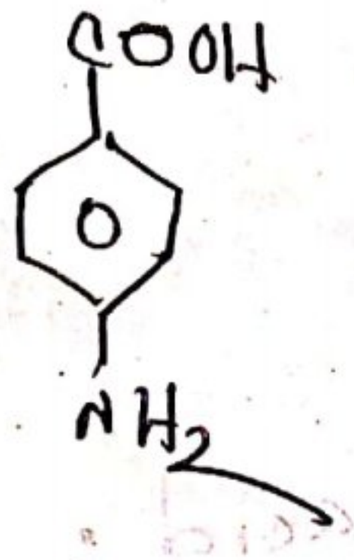
* 4-Hydroxy benzaldehyde:



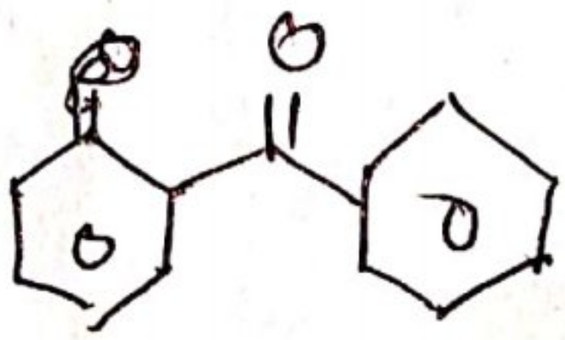
* Benzamide:



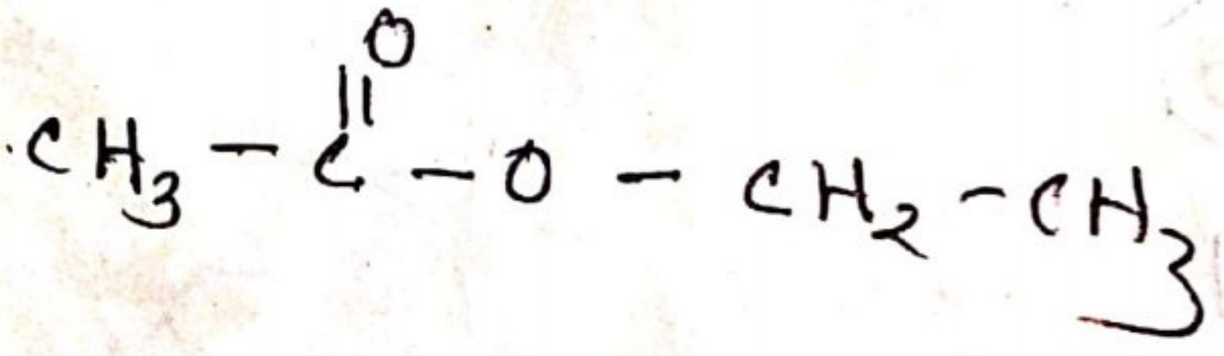
* 4-aminobenzoic acid:



* Benzophenon.



* Ethylacetate

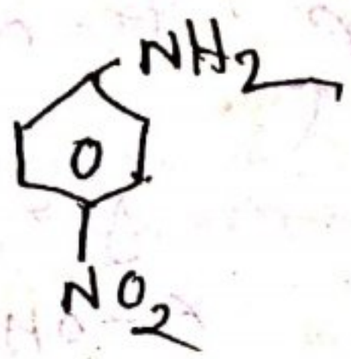


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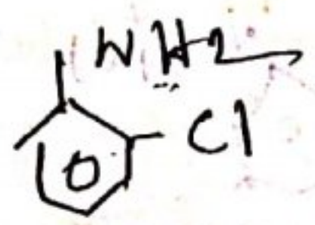
* Acetanilide



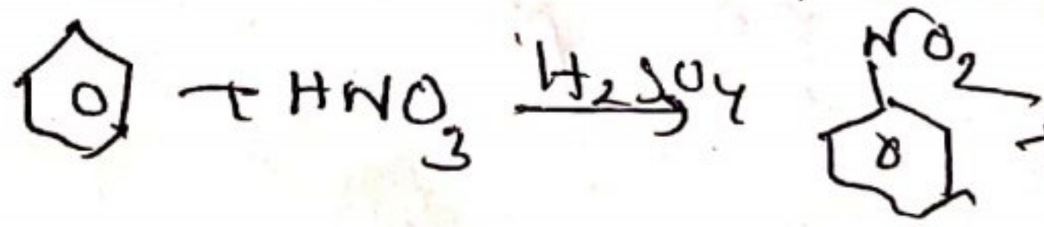
* 4-nitro aniline:

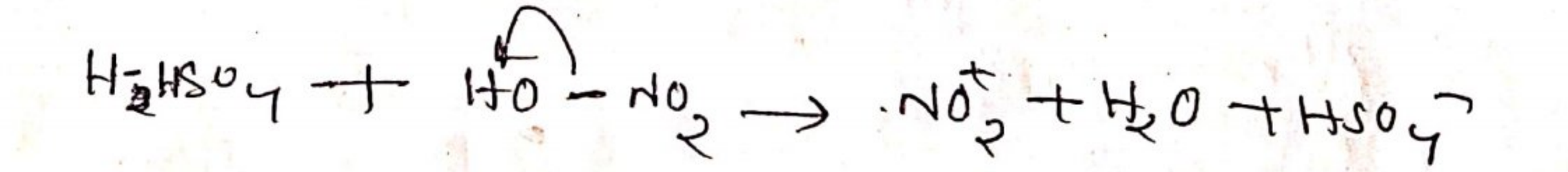


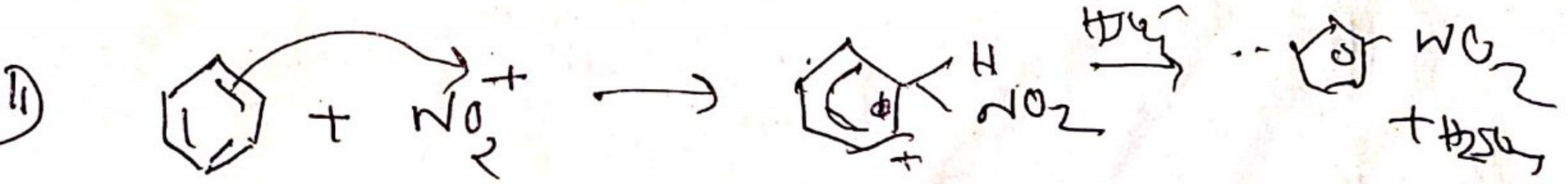
* 2-chloro aniline



* mechanism of Nitration:



1) 

2) 

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Tollen's reagent preparation :-

. 90 ml $AgNO_3$ + 1 drop 10% NaOH + NH_3 solution যতক্ষণ না পর্যন্ত মিশ্রণটি clear হয় ।

$$AgNO_3 + NaOH \longrightarrow AgOH + NaNO_3$$

$$\downarrow$$

$$Ag_2O$$

$$Ag_2O + NH_4OH \longrightarrow Ag(NH_3)_2OH$$

tollen's reagent

$$R-CHO + Ag(NH_3)_2OH \longrightarrow Ag\downarrow + NH_3 + RCOONH_4 + H_2O$$

$$Na_2S + Na_2[Fe(CN)_5NO] \longrightarrow Na_4[Fe(CN)_5NOS]$$

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Aldehide and ketone:

$Ag(NH_3)_2OH$ → oxydizing agent

$-\overset{O}{\overset{\|}{C}}-H$ → oxidize করে → $COOH$

-এখানে $Ag(NH_3)_2OH$ mild oxidizing agent তাই কিটোনকে oxidize করতে পারে না।

$$R-\overset{O}{\overset{\|}{C}}-R$$

কিন্তু Heat দিলে $Ag(NH_3)_2OH$ কেটে গিয়ে $Ag\downarrow$ পড়ে। ফলে reaction হয় মাত্র মে $Ag\downarrow$ পড়ে না।

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Q: Why run solubility test before running the functional group classification test? Determine the functional group present in these unknown?

(a) unknown A is soluble in water and gives bubbles with 5% $NaHCO_3$.

(b) unknown B is insoluble in water insoluble in 5% NaOH but soluble in 5% HCl.

(c) unknown C is insoluble in water insoluble in 5% NaOH, insoluble in 5% HCl, soluble with a color change in conc. H_2SO_4 and decolorizes both KMnO4 and Bromin in CCl_4.

Ans:

After analysis for the element present, the next important step is the determination of solubility of the unknown organic compounds to be tested. It provides with many informations as regards the nature of the compounds. These information

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are very much helpful prior to the specific test for functional groups.

(i) The presence of a functional group is often indicated. For instance, because hydrocarbons are insoluble in water, the mere fact that an unknown compound (such as diethyl ether) is partially soluble in water indicates that a polar functional group is present in the unknown molecule.

(ii) Solubility in certain solvents often leads to more specific information about the functional group. For example, benzoic acid is insoluble in polar solvent, water but is converted by 5% NaOH solution to a salt, sodium benzoate, which is readily water soluble. In this

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case, then solubility in 5% NaOH solution of a water insoluble unknown organic compound is a strong indication of an acidic functional group.

For this reason, solubility test run before running the functional group classification test.

(b) unknown B is insoluble in water, insoluble in 5% NaOH but soluble in 5% HCl.

Solvent	H_2O	5% NaOH	5% NaHCO3	5% HCl	Conc. H_2SO_4	85% H_3PO_4
test	Insoluble	Insoluble	—	soluble		

class B; basic compounds:

class B: Aliphatic amines with more than eight or more carbon atoms; aromatic amines such as anilines (only one phenyl group attached with nitrogen); some ethers.

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(C) unknown C is insoluble in water, insoluble in 5% NaOH, insoluble in 5% HCl, soluble with a color change in conc. H_2SO_4 and decolorizes both $KMnO_4$ and Bromin in CCl_4

Solvent	H_2O	5% NaOH	5% $NaHCO_3$	5% HCl	Conc. H_2SO_4	85% H_3PO_4
Test	Insoluble	Insoluble	—	Insoluble	Soluble	—

decolorize both $KMnO_4$ and Br_2 in CCl_4

Class - N_2

unsaturated hydrocarbons: alkenes, alkynes, ethers, ketones.

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How is Tollen's reagent prepared?

It is prepared using a two step procedure.

Step-1: ~~Aqous~~ Aqueous Silver nitrate is mixed with aqueous sodium hydroxide.

Step-2: Aqueous ammonia is added drop wise until the precipitated silver oxide completly dissolves.

Tollen's reagent oxidizes an aldehyde into the corresponding carboxylic acid.

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Q: what is sodium fusion test ?

The sodium fusion test is used in elemental analysis for determination of elemental halogens, nitrogen and sulfur in a sample. An organic compound is heated with metallic sodium using extensive decomposition, carbon is deposited and some CO and CO_2 are formed. Nitrogen, in the presence of carbon and sodium forms cyanide (sodium cyanide, NaCN): sulphur forms sulphide (sodium sulphide, Na_2S) and halogen form the corresponding halides (sodium halides, NaX ; X = Cl, Br, I)

organic compounds containing C, N, S, X

$\xrightarrow{Na\ +\ Heat}$ $Na_2S + NaCN + NaCNS + NaX\ [X = Cl, Br, I]$

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NaCNS will form only then when metallic sodium is available less than required.

Q: Why KF is added in N identification 2

The addition of the minute amount of KF facilitates the conversion of extra Fe^{3+} ion to $K_3[FeF_6]$ (potassium fennifluonide) which leads to the formation of a <u>purely</u> Prussian blue.

$$KI + Fe^{3+} \rightarrow K_3[FeF_6]$$

Q: Why HCl was not used in acidifying the alkaline solution 2 (N identification)

we usually use dil. H_2SO_4 but on HCl must be avoided on otherwise $FeCl_3$ can be formed which is of yellow colour and can cause the Prussian blue to appear greenish.

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why H_3PO_4 is used when after the addition of dil. H_2SO_4 the colour of $Fe_4[Fe(CN)_6]_3$ appears greenish ? (N detection)

This existence of greenish colour usually occurs due to the presence of persisting Fe^{3+} ion which is then immediately converted to a colourless watersoluble complex $[Fe(HPO_4)^+ Cl]$. Then the prussian blue appears if N is present.

why sodium metal is used in Lassaigne test ?

Na metal is very reactive, that is covalently bonded N, S, X etc in organic compounds can form ionic compounds by reacting with Na, and produced ionic sodium salts are soluble in water.

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Give the detection of sulphur :

Firstly the fusion filtrate is acidified with dil. CH_3COOH until the solution is acidic. Then if we add lead acitate, the solution containing the sulphur reacts to give the black precipitate of PbS which indicates the presence of sulphur.

$$Na_2S + (CH_3COO)_2Pb \xrightarrow{dil.\ CH_3COOH} PbS\downarrow + 2CH_3COONa$$

Q: why do we need to acidify the solution with dilute acetic acid ?

During the fusion process, if any trace of sodium metal remains unreacted, it will form NaOH with water, in that case NaOH may react with $Pb(CH_3COO)_2$ to give the black precipitate of $Pb(OH)_2$ which may hesitate us.

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So we need to add dil. CH_3COOH to neutralize the NaOH if present

$$NaOH + CH_3COOH \longrightarrow CH_3COONa + H_2O$$

Sodium nitroprusside test: If we add only a drop of 0.10% sodium nitroprusside to our ~~fusion~~ fusion filtrate if the filtrate contains sulphur, the reaction will produce a deep red or purple colour solution

$$Na_2S\ (aq) + Na_2[Fe(CN)_5NO]\ (aq) \longrightarrow$$

Sodium nitroprusside

$Na_4[Fe(CN)_5NOS]$ aq.
A deep red or purple colour.

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Q. How would you give a combine test for N and S ?

If any organic compound contains N and S, during the sodium fusion it may produce sodium sulfocyanide or sodium thiocyanate (NaCNS). A fusion filtrate containing NaCNS is first acidified with dil. HCl Then a drop of freshly prepared 2.5 % $FeCl_3$ will give a blood red colonation due to the formation of the compound ferric thiocyanochloride which indicates the presence of N and S.

$$FeCl_3 + NaCNS \rightleftharpoons Fe(CNS)Cl_2$$

Sodium sulfocyanide — Ferric thiocyano-chloride (blood red)

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a: what would happen when sodium be deficient in fusion test ?

gt is essential to use an excess of sodium otherwise if sulphur and nitrogen are both present sodium thiocyanate NaSCN, may be produced in the test for nitrogen it may give a red colonation with ferric ions but no prussian blue since there will be no free cyanide ions. with excess of sodium the thiocyanate will be decomposed :-

$$NaSCN + 2Na \longrightarrow NaCN + Na_2S$$

detection of nitrogen with reactions :—

$$NaCN + FeSO_4 \rightarrow Na_4[Fe(CN)_6] + Na_2SO_4$$

$$Na_4[Fe(CN)_6] + FeCl_3 \rightarrow$$

$$Fe_4[Fe(CN)_6]_3$$

Prussian blue

Ferric ferro cyanide

$Fe_4[Fe(CN)_6]_3$

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Q: why methanol is used in Lassaigne test ?

After heating the composition of sodium and sample, it is cooled to room temperature, then we add 5-7 drops of methanol before adding water. If the composition after being cooled contains any sodium, it is removed as $CH_3O^-Na^+$ by adding CH_3OH. Otherwise if we don't add CH_3OH before using H_2O, Na will react vigorously with H_2O to result damage.

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Q: why cyanide or sulphide ions must removed before adding silver ions to test for the halides ?

The Lassaigne sample, when selected to test for hallide ions, it is obvious to remove cyanide and sulphide ions, if any in sample.

$$AgNO_3 + CN^- \longrightarrow AgCN \downarrow + NO_3^-$$

precipitate

$$AgNO_3 + S^{2-} \longrightarrow Ag_2S \downarrow + NO_3^-$$

precipitate

we see that if CN^- and S^{2-} ions are not removed, they will react with Ag^+ to result in positive hallide test though the sample does not really contain any hallogen.

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Q: How to remove CN^- and S^{2-} ions?

we can remove CN^- and S^{2-} ions from the sample by adding conc. HNO_3 acid to it.

$$NaCN + HNO_3 \xrightarrow{\Delta} NaNO_3 + HCN$$

$$Na_2S + HNO_3 \xrightarrow{\Delta} NaNO_3 + H_2S$$

. $NaNO_3$ being a soluble salt; does not interfere further.

N.B HCN এবং H_2S বিষাক্ত গ্যাস।

Q: why the sample after being acidified with conc. HNO_3 need to boiled.

The removal of HCN/H_2S made from the reaction of CN^- and S^{2-} ions with HNO_3 is effected by boiling the fusion solution to reduce it to half of its initial volume.

The sample after being acidified with conc. HNO_3 need to boil to remove HCN and H_2S.

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N.B HNO3 ছাড়া অন্য কোন acid use করা যাবে না তাহলে precipitate পড়বে। H2SO4 use করলে Na2SO4 সাদা precipitate পড়বে।

Q: How would you differentiate chloride, bromide and iodide in the fusion extract?

After being assumed about the presence of halide ions; then we would take 0.50 ml of the fusion filtrate + 0.50 ml CCl_4 + 1.0 ml conc. HNO_3. After shaking the mixture we would allow the liquid of the test tube to statify

$NaCl \xrightarrow[CCl_4]{HNO_3} NaNO_3 + HCl$; $HCl + HNO_3 \longrightarrow$ Cl_2 (in CCl_4) colourless

$NaBr \xrightarrow[CCl_4]{HNO_3} NaNO_3 + HBr$; $HBr + HNO_3 \rightarrow$ Br_2 in CCl_4 Deep brown/ tan red

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$$NaI \xrightarrow[CCl_4]{HNO_3} NaNO_3 + HI \;;\; HI + HNO_3 \rightarrow I_2$$

(in CCl_4)

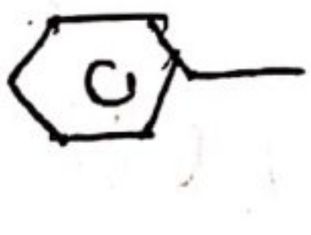
violet

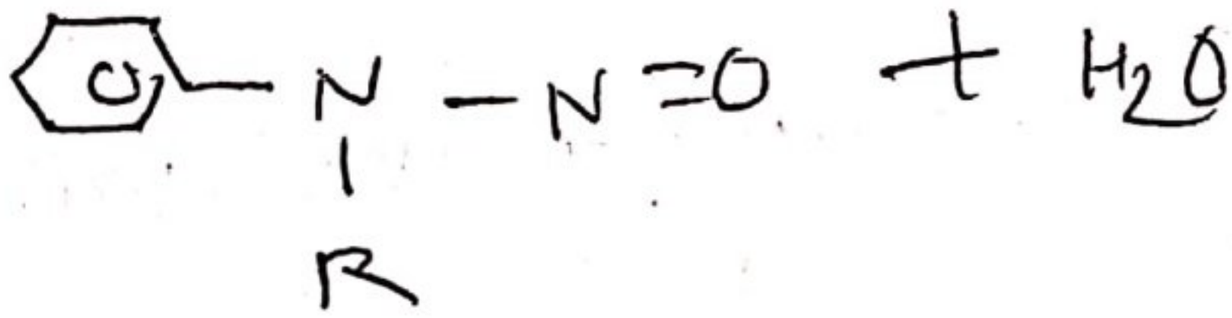
The different colounes of molecule bromine, iodine and chlorine indicates the presence of Cl^-, Br^-, I^- ions in the fusion filtrate.

Test for secondary amine

① $R_2NH + HNO_2 \longrightarrow R_2-N-NO + H_2O$

oily dank colon liquid

①  $+ HNO_2 \longrightarrow$



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② Libermann's test :-

① $HCl + NaNO_2 \rightarrow HNO_2 + NaCl$

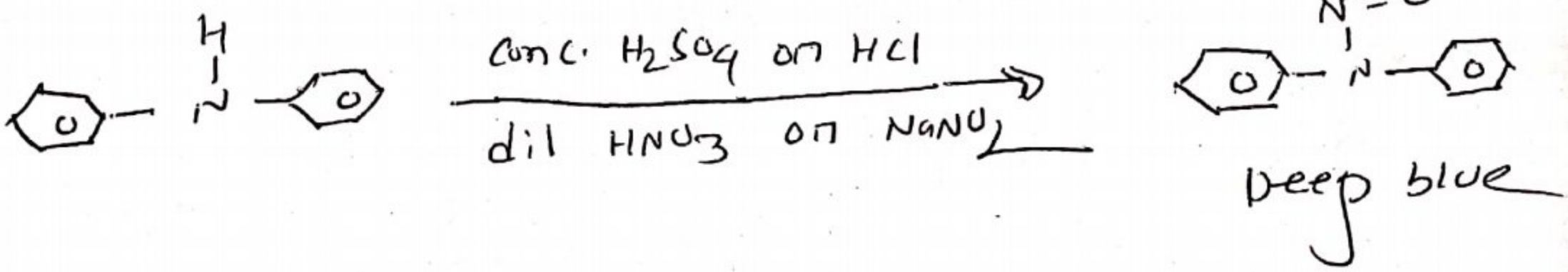
$R_2NH + HONO \longrightarrow R_2-N-N=O + H_2O$

$R_2-N-N=O \xrightarrow[\text{conc. } H_2SO_4]{\text{phenol}}$ blue colouration

$\downarrow$ NaOH (5%)

Blue black

(3) specific test for Diphenyl amine :



Test for tertiary amine :

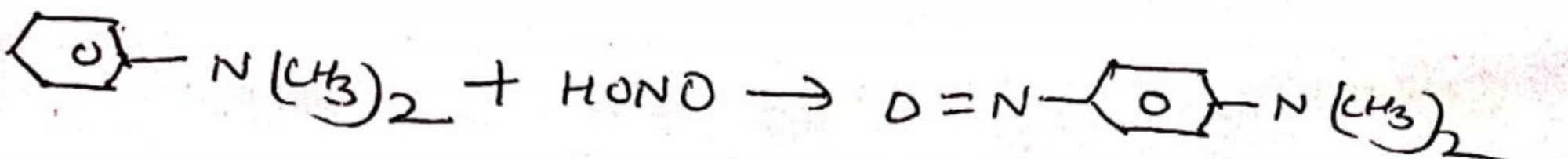
① Nitrous acid test : aliphatic

$(CH_3)_3N + HO-NO \rightarrow [(CH_3)_3N^+H]\ NO_2^-$

[No reaction]

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Anomatic:



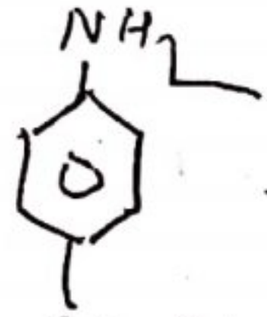
p-nitno dimethyl aniline (green)

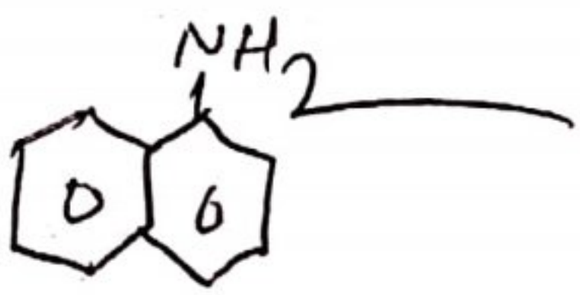
*** no reaction indicates aliphatic tertiary amine.

*** production of green and brown colored salt indicates the presence of anomatic tertiary amines.

Test for nitno compounds:-

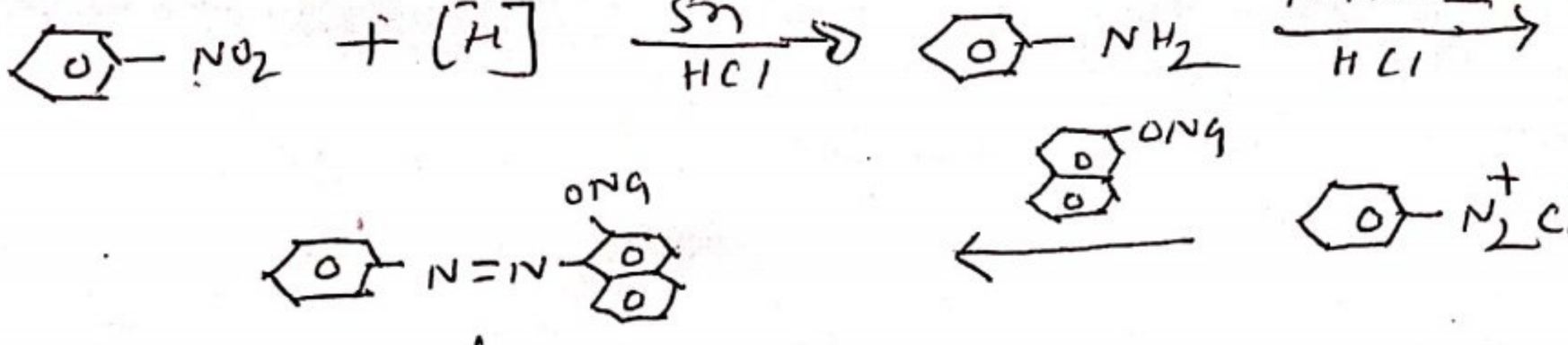
Test with griess reagent:

equal volume of 1% 

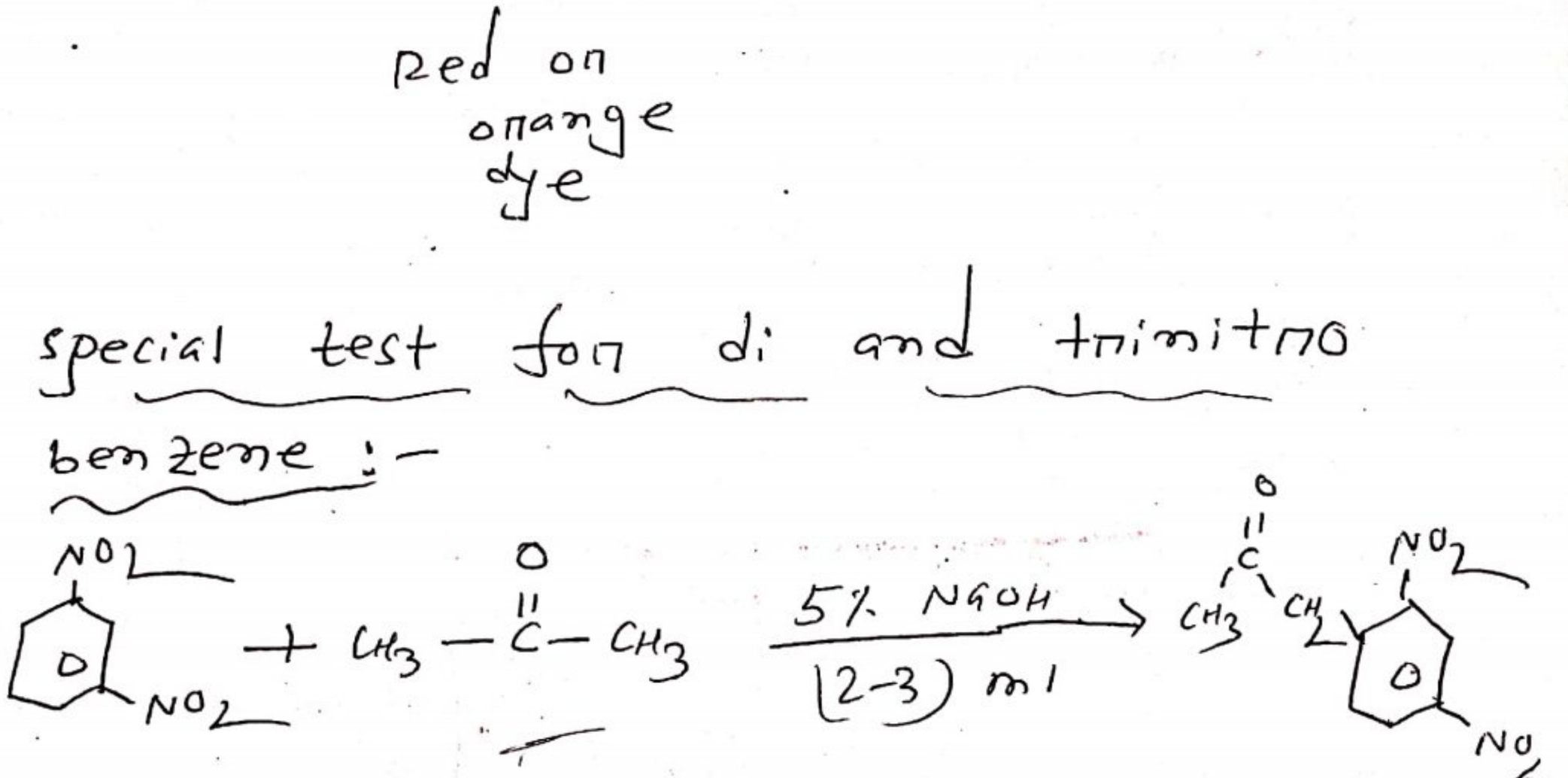
and 0.1% 1 napthyl amin 

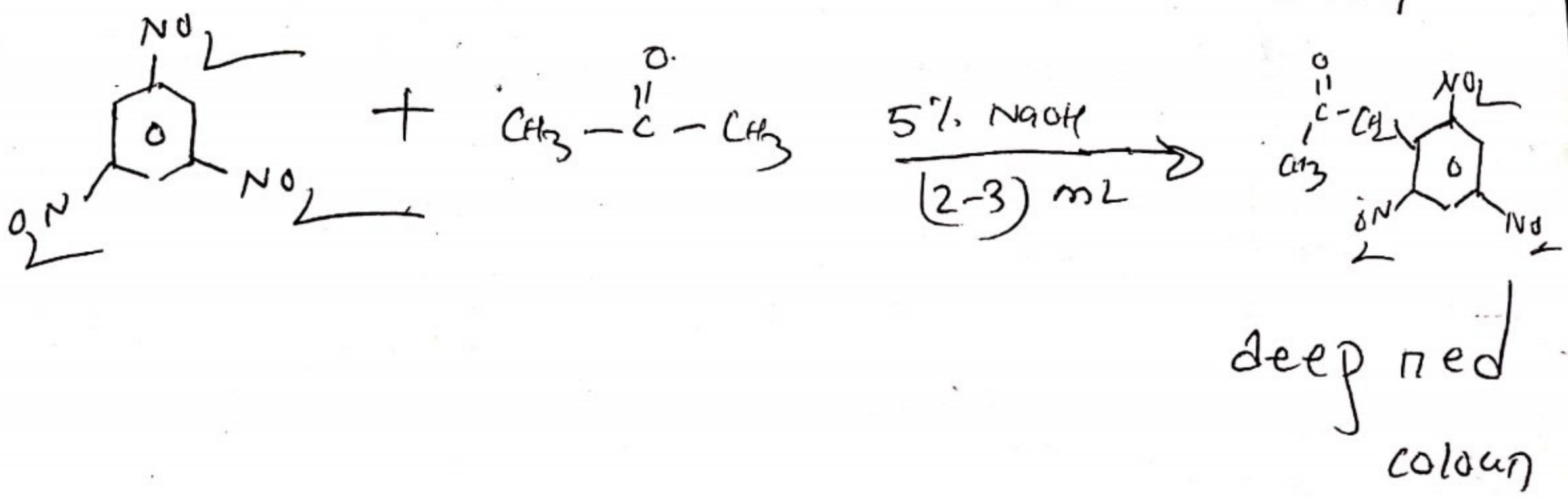
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(11) Reduction of nitro compound:



special test for di and trinitro benzene :-





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General test for amide :

① $R-CO-NH_2 + HNO_2 \xrightarrow[HCl\ (0-5^0C)]{NaNO_2} R-\overset{O}{\overset{\|}{C}}-OH + N_2 + H_2$

Brisk effervescene due to evolution of N_2 gas

② NaOH test :

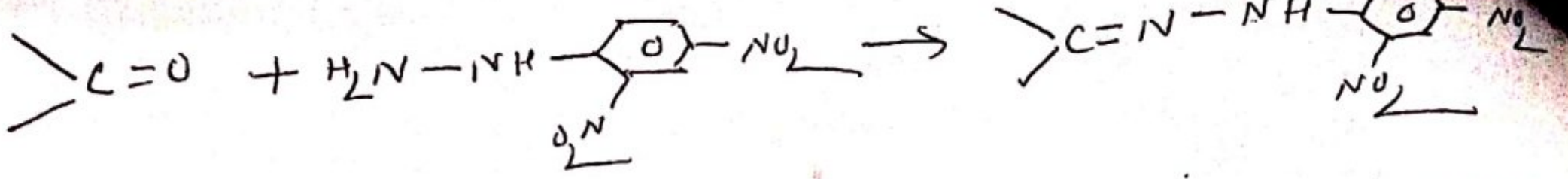
$R-\overset{O}{\overset{\|}{C}}-NH_2 + NaOH \longrightarrow RCOO-Na + NH_3$

$NH_3 + CuSO_4 \longrightarrow [Cu(NH_3)_4]SO_4$

Blue Colon.

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Test for carbonyl compounds



Colon precipitate

Test for aldehydes :

① Reaction with schiff's reagent :

$$-CHO + SOCl_2 \longrightarrow -C(=O)-Cl + SO_2$$

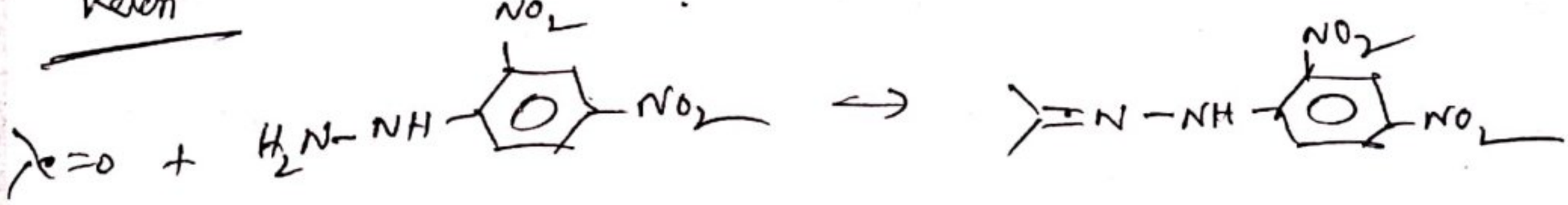
wine purple

Tollen's test :

$$R-\overset{O}{\overset{\|}{C}}-H + 2Ag(NH_3)_2OH \longrightarrow 2Ag\downarrow + 4NH_3 + RCOONH_4$$

Mirror

Keton



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mechanism :

$$AgNO_3 + NaOH \rightarrow AgOH + NaNO_2$$

$$\downarrow$$

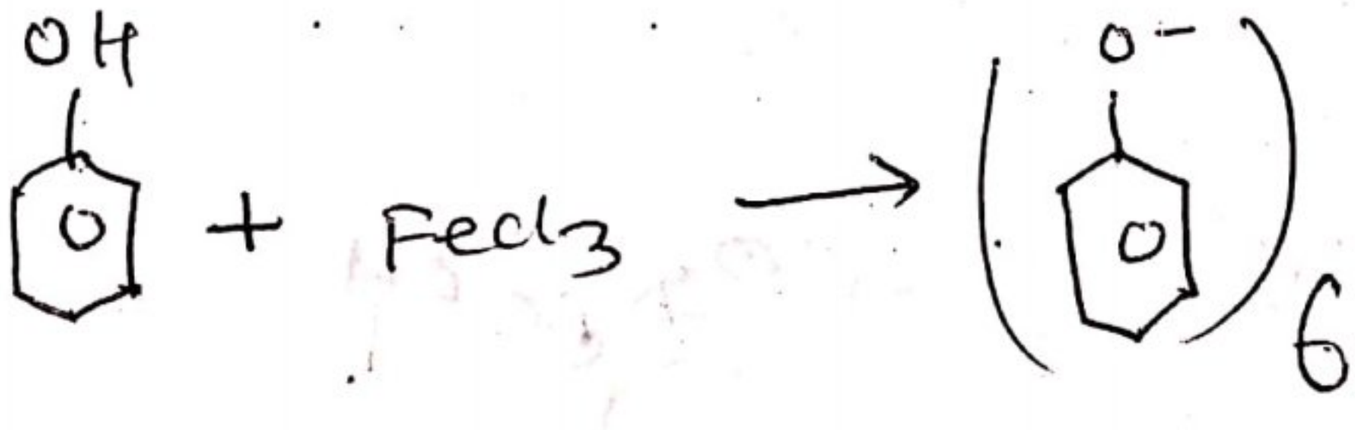
$$Ag_2O$$

$$Ag_2O + NH_4OH \longrightarrow Ag(NH_3)_2OH$$

↑ Tollen's reagent

$$R-CHO + Ag(NH_3)_2OH \rightarrow 2Ag\downarrow + NH_3 + RCOONH_4 + H_2O$$

Test for phenol :



$$+ FeCl_3 \longrightarrow$$

$$_6 \quad Fe_2 + HCl$$

$$C_6H_5OH + FeCl_3 \rightarrow (C_6H_5O)_6 Fe_2 + HCl$$

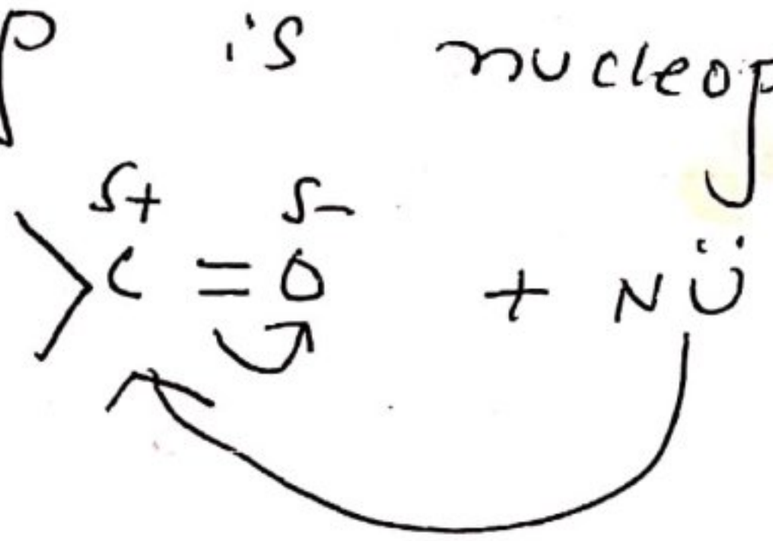
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* test for carboxylic acid :

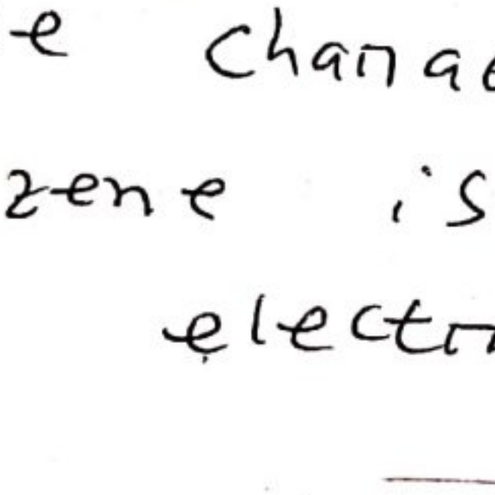
$R-COOH + NaHCO_3 \longrightarrow RCOONa + CO_2 \downarrow + H_2O$

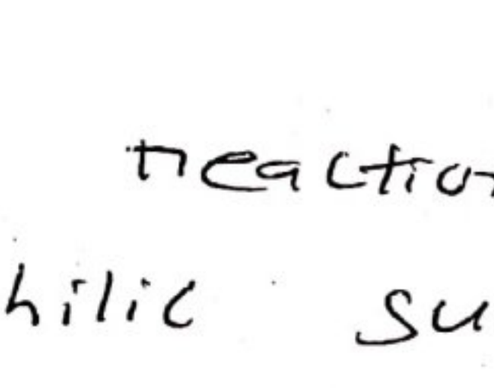
CO_2 এর বুদবুদ হয়।

* The characteristic reaction of carbonyl group is nucleophilic addition.



* The characteristic reaction of benzene is nucleophilic substitution and electrophilic addition.





$R-O-R$

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Jahangirnagar University

Organic Chemistry Laboratory

Course Title-Identification of Organic Compounds, Course No-335

IR Data

Typical Range/ cm^{-1}	Functional groups	Bond suspected	Descriptions
3640-3610	Alcohols, Phenols	O-H stretch(Free OH)	S, Sh
3600-3200	Alcohols, Phenols	O-H stretch(H-bonded)	S, b
3600-2500	Carboxylic acid	O-H stretch(H-bonded)	s
3400-3200	Amine	N-H Stretch	m
3100-3000	Aromatics Alkene	=**C-H** stretch	s
3000-2850	Alkane	C-H stretch	m
2900-2700	Aldehyde	O=**C-H** stretch	w
2280-2230	Nitrile	C=N stretch	m
2260-2200	Alkyne	C=C stretch	m
1850-1800 1790-1740	Anhydride	C=O stretch Usually two bands	s
1760-1690	Aldehyde, ketone, amide, ester	C=O stretch	s, m
1690-1650	Imine	C=N stretch	v
1600-1500	Alkene	C=C stretch	m
1600-1515	Nitro	N=O stretch	s
1470-1450	Alkane	C-H bend	m
1320-1060	Alcohol, carboxylic acid, ester, ether	C-O stretch	s
1000-650	Alkene	=C-H bend	m
950-910	Carboxylic acid	O-H bend	m
1400-1000	Halide	C-F stretch	w
800-600	Halide	C-Cl stretch	s
650-500	Halide	C-Br stretch	s

****

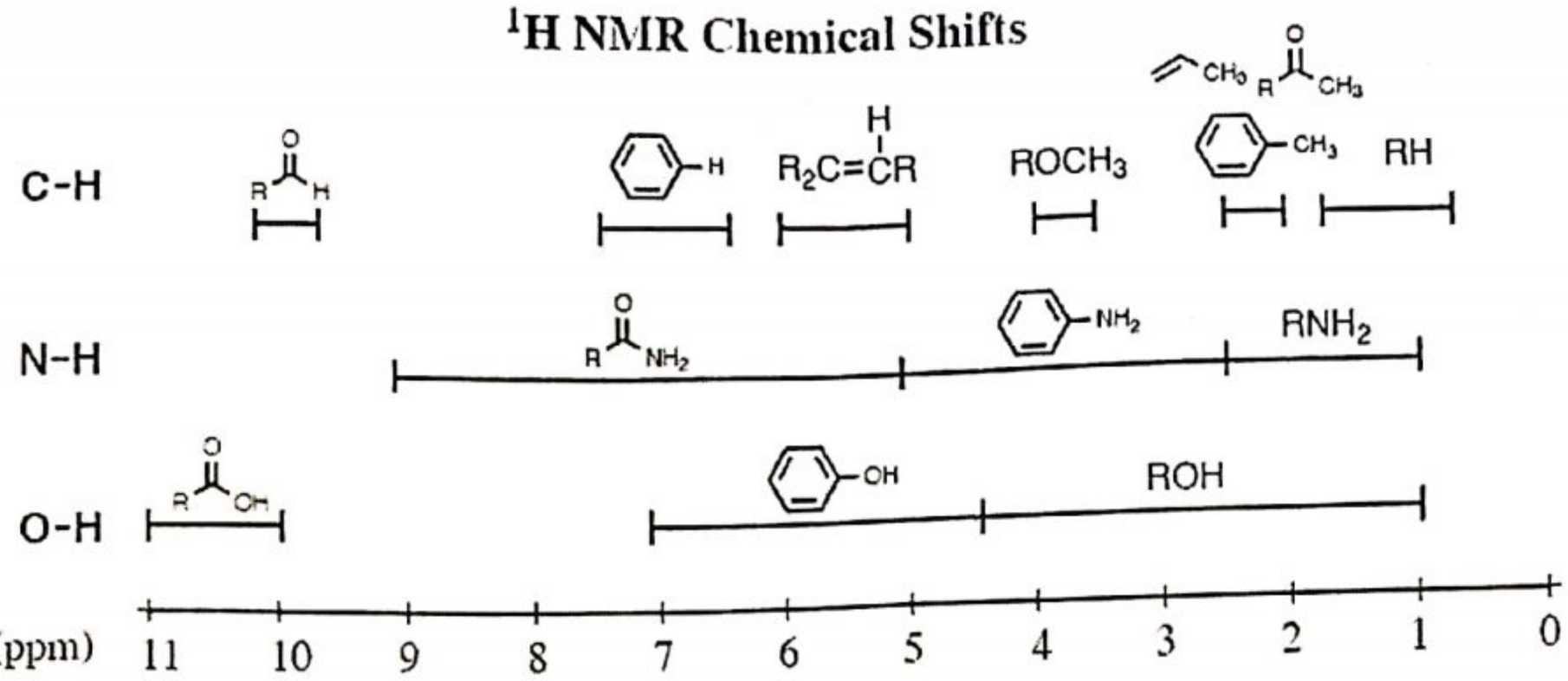
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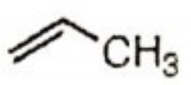
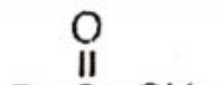
Organic Chemistry Laboratory

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^{1}H NMR Chemical Shifts

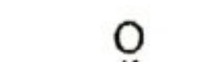


Protons on Carbon

Type of C-H	δ (ppm)	Description of Proton
$R-CH_3$	0.9	alkyl (methyl)
$R-CH_2-R$	1.3	alkyl (methylene)
R_3C-H	1.5-2	alkyl (methine)
	1.8	allylic (C is next to a pi bond)
	2-2.3	α to carbonyl (C is next to C=O)
$Ar-CH_3$	2.3	benzylic (C is next to Ph)
$RC\equiv C-H$	2.5	alkynyl
R_2N-CH_3	2-3	α to nitrogen (C is attached to N)
$R-CH_2-X$	2-4	α to halogen (C is attached to Cl, Br, I)
$RO-CH_3$	3.8	α to oxygen (C is attached to O)
$R-CH_2-F$	4.5	α to fluorine (C is attached to F)
	5-5.3	vinylic (H is attached to alkene C)
$Ar-H$	7.3	aromatic (H is on phenyl ring)
	9.7	aldehyde (H is on C=O)

Note: aldehyde (-CHO) proton usually does not couple with neighboring H's so appears as a singlet

Protons on Oxygen/Nitrogen*

Type of H	δ (ppm)	Description
ROH	0.5-5	alcohol
ArOH	4-7	phenol
	10-13	carb. acid
RNH_2	0.5-5	amine
$ArNH_2$	3-5	aniline
	5-9	amide

**Protons on N or O typically have wide ranges of expected chemical shifts; the actual δ value depends on the solvent used, the concentration, temperature, etc. Because these protons are acidic and, therefore, exchangeable, they may be broad peaks and usually do not couple with neighboring protons (typically they are broad singlets). If a protic deuterated solvent is used (e.g., D_2O or CD_3OD), then the NH and OH protons will exchange with the deuterium and the peaks will shrink or disappear entirely, since D (^2H) does not show up in the ^1H NMR spectrum.*

R = alkyl group
Ar = aromatic ring, such as phenyl (Ph)

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